

Evaluating Belief Formation and Belief–Action Coupling in Large Language Models

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Abstract

Intelligent behavior under uncertainty demands more than good outcomes—it demands that actions be *driven* by beliefs. A rational agent facing a partially observable environment must infer latent structure from observations, represent that structure as a calibrated probability distribution, and select actions that are optimal under it. Whether large language models (LLMs) implement this pipeline—or merely produce behavior that correlates with it—is a question that existing benchmarks cannot answer. Payoff maximization and task success, the dominant metrics in LLM evaluation, are consistent with both genuine belief-driven reasoning and sophisticated pattern-matching that happens to yield similar outcomes under standard conditions.

We argue that the core question is causal: do an LLM’s stated beliefs *cause* its actions, or are beliefs and actions parallel outputs of a common heuristic, independent of each other? Answering this requires a setting where the two can be dissociated—where we can hold the environment fixed while manipulating beliefs, and measure whether actions follow. Repeated strategic interaction with hidden opponent types provides exactly this structure. The opponent’s latent policy creates a ground-truth inference problem with a computable normative reference; stated beliefs can be elicited explicitly; and beliefs can be causally manipulated by injecting counterfactual posteriors while the game state remains unchanged.

We introduce a benchmark that operationalizes this causal decomposition across four canonical 2×2 games and nine opponent policy types, spanning deterministic, stochastic, and deceptive behavioral strategies. Our framework separates strategic reasoning into four measurable components—latent inference, uncertainty calibration, belief–action coupling, and temporal belief dynamics—and evaluates each against a Bayesian oracle posterior derived from the known opponent policy definitions.

Our central finding is a systematic decoupling between stated beliefs and realized actions: models frequently report posteriors that prescribe one EU-optimal action while selecting another, and respond to injected belief changes in ways that are sensitive but not directionally rational. Strong behavioral performance does not imply coherent belief-driven reasoning. The benchmark provides the diagnostic infrastructure to distinguish these failure modes—separating models that reason from models that approximate reasoning—and establishes a foundation for building LLM agents whose decisions are genuinely grounded in what they infer.

1 Introduction

The gap between behavior and reasoning. Large language models are increasingly deployed as autonomous agents in interactive environments, where they must make sequences of decisions under uncertainty [Akata et al., 2025, Duan et al., 2024]. Performance in these settings is almost universally measured by observed outcomes: cumulative reward, task success rate, equilibrium adherence. These metrics are natural and practically important, but they carry a fundamental ambiguity—they cannot distinguish between an agent that *reasons* about the environment and an agent that produces behavior correlated with good reasoning through pattern-matching heuristics. In partially observable settings, this distinction matters enormously. An agent whose actions are genuinely conditioned on its beliefs will generalize to novel environments, respond predictably to new information, and fail in diagnosable ways. An agent that emulates belief-driven behavior through surface heuristics will not.

What the question really is. Effective decision-making under partial observability requires three separable components to work in concert. The agent must first *infer* latent structure—in our setting, the hidden policy of an opponent—from observed interactions. It must then *represent* that inference as a calibrated probability distribution over the latent variable space. And it must finally *condition its actions* on that distribution in a manner consistent with expected utility maximization. Existing evaluations conflate all three. A model that achieves high payoff may be doing so because it infers accurately, or because it calibrates well, or because a game-structure heuristic happens to select good actions independently of any belief. Separating these explanations requires not just measuring outcomes but decomposing the inference–representation–action pipeline and testing each link in isolation.

The question we pose is causal, not correlational: do an LLM’s stated beliefs *cause* its actions? Correlation between beliefs and actions is a necessary but not sufficient condition for the answer to be yes. A model that consistently cooperates against cooperative opponents may be acting rationally under its beliefs, or it may simply be applying a cooperation heuristic that coincidentally aligns with the belief-optimal action. These hypotheses are observationally equivalent under standard evaluation conditions and require deliberate intervention to separate.

Why strategic interaction? Strategic repeated interaction with a hidden opponent provides a uniquely clean setting for this investigation. The opponent’s latent type creates a well-defined inference problem with a computable normative solution—the Bayesian posterior over policy types given the observed interaction history. This oracle posterior serves as a ground-truth reference against which elicited beliefs can be scored, without requiring access to the model’s internal states. At the same time, different opponent types induce qualitatively different EU-optimal actions under different beliefs, creating natural conditions under which the belief–action link can be tested and, more importantly, *broken* by injecting counterfactual posteriors while the game state remains fixed. The opponent’s hidden type thus acts as the latent variable that makes the entire causal structure testable.

The diversity of the four games—Prisoner’s Dilemma, Stag Hunt, Chicken, and Battle of the Sexes—is not incidental. Different payoff structures invert the EU-optimal action under different beliefs: cooperation is dominant in Stag Hunt under beliefs favoring reciprocal types, while it is dominated in the Prisoner’s Dilemma regardless of beliefs. If a model’s cooperation rate is insensitive to this variation, its actions are being driven by game-structure heuristics rather than inferred opponent type—a form of belief–action decoupling visible only when multiple incentive regimes are compared simultaneously.

The intuition behind our approach. Consider an LLM that, at round 6 of a Prisoner’s Dilemma, reports a posterior heavily concentrated on `tit_for_tat`—an opponent who mirrors the agent’s last action. Under this belief, the EU-optimal action is unambiguously *C*: if the agent cooperated last round, the opponent will cooperate, and mutual cooperation yields the highest symmetric payoff. Yet the model plays *D*. This single observation is not an anomaly in our results—it is the modal pattern. Models consistently articulate beliefs that prescribe one action and select another. The stated posterior and the action are two outputs of the same forward pass that are largely independent of each other. This is what we call *belief–action decoupling*, and it is the central empirical phenomenon this paper characterizes, measures, and diagnoses.

The insight that drives our framework is that decoupling can be measured at multiple levels of causal strength. At the weakest level, we ask whether the model’s action is *sensitive* to changes in the injected posterior at all. At a stronger level, we ask whether changes are in the *correct direction* as prescribed by expected utility. At the strongest level, we ask whether the model’s own stated posterior—not an externally injected one—is the actual driver of its action selection. These three levels form a diagnostic hierarchy: a model can pass lower levels while failing higher ones, and the pattern of failures precisely locates the breakdown in the pipeline.

Contributions. This paper makes the following contributions:

1. We introduce a benchmark for evaluating belief dynamics and belief-conditioned action in LLMs, instantiated across four canonical 2×2 games and nine opponent policy types spanning deterministic, stochastic, and deceptive behavioral strategies.
2. We formalize the decomposition of strategic reasoning into four measurable components—latent inference, uncertainty calibration, belief–action coupling, and causal posterior sensitivity—each evaluated against an oracle Bayesian reference derived from known opponent policy definitions.
3. We introduce a suite of preregistered metrics that operationalize this decomposition, including EU-consistent coherence, directional accuracy under posterior intervention, surprisal–update coupling, and a theory-of-mind gap metric targeting deceptive opponent identification.
4. We propose a causal evaluation protocol—the Posterior Intervention experiment—that injects counterfactual belief distributions while holding the game state fixed, enabling behavioral-level causal inference about the belief–action link without access to model internals.
5. We provide a diagnostic taxonomy of failure modes separating inference errors, calibration failures, coupling failures, and decision-policy errors, enabling fine-grained characterization of where in the pipeline a model breaks down.

Key findings. Across six models spanning five provider families, we find: EU-consistent coherence—the fraction of probe-round actions matching the EU-optimal action under the model’s own stated posterior—averages near chance across all games, revealing that stated beliefs and actions are largely decoupled processes. Directional accuracy under posterior intervention is substantially below 1.0, and the gap between sensitivity and directional accuracy exposes widespread irrational responsiveness: models react to belief changes but not in the direction expected utility prescribes. Rigidity indices are high, indicating that cooperation rates are largely invariant to game structure, consistent with heuristic rather than belief-conditioned behavior. Together, these results establish that strong behavioral performance is neither necessary nor sufficient evidence for coherent belief-driven reasoning in LLMs.

Paper organization. Section 2 situates the work in related literature. Section 3 formalizes the problem setup and notation. Section 4 describes the benchmark framework and its five experimental stages. Section 5 defines the full metric suite. Section 6 describes the experimental setup and implementation. Section 7 presents empirical results. Section 8 discusses implications and limitations. Section 9 concludes.

2 Related Work

2.1 Strategic and Game-Theoretic Evaluation of LLMs

The closest body of related work evaluates LLM behavior in game-theoretic settings. Akata et al. [2025] introduced the systematic study of LLMs in finitely repeated 2×2 games—including the Prisoner’s Dilemma and Battle of the Sexes—against human-like strategies, finding that GPT-4 is particularly unforgiving in the Prisoner’s Dilemma but fails to coordinate in games that require alternation. This work was later published and extended in *Nature Human Behaviour* [Akata et al., 2025], establishing that behavioral signatures are stable across robustness checks and can be modulated via social chain-of-thought. Duan et al. [2024] introduced GTBench (NeurIPS 2024), a language-driven benchmark across 10 game-theoretic tasks spanning complete/incomplete information and deterministic/probabilistic settings, finding that LLMs fail in complete deterministic games while remaining competitive in probabilistic ones. Costarelli et al. [2024] extended this to out-of-distribution games with sparse pretraining coverage. Phelps and Russell [2025] conducted the first evolutionary Iterated Prisoner’s Dilemma tournaments with frontier LLMs, finding persistent model-specific strategic fingerprints.

A key limitation shared across all these works is that they measure *behavioral outcomes*—payoffs, cooperation rates, equilibrium adherence—without probing whether observed behavior is driven by inferred beliefs about the opponent. Our framework addresses this gap directly by eliciting explicit posterior distributions and testing their causal role in action selection.

2.2 LLMs as Decision-Making Agents under Uncertainty

Binz and Schulz [2023b] studied GPT-3 through cognitive psychology tools, finding signatures of model-based reasoning alongside systematic failures in causal reasoning and directed exploration. Most recently, Freedman et al. [2026] tested whether LLM elicited probabilities satisfy coherence conditions consistent with rational utility maximisation in medical diagnostic settings — the closest prior work to our own. Critically, their setting involves static factual uncertainty with no opponent, no repeated interaction, and no oracle reference posterior; they test axiomatic coherence conditions rather than causal belief–action coupling via counterfactual injection, and do not instrument the inference–representation–action pipeline. Binz and Schulz [2023a] demonstrated that fine-tuning LLMs on behavioral datasets can produce accurate cognitive models that outperform traditional decision-making models. These works establish that LLMs can be studied as decision-making agents using tools from cognitive science—a perspective our work extends to the strategic, partially observable setting. Horton et al. [2023] examined LLM agents in economic and strategic contexts, evaluating consistency with expected utility maximization. However, none of these works decompose the inference–representation–action pipeline or test whether beliefs causally drive decisions.

2.3 Calibration and Uncertainty in LLMs

A substantial literature evaluates LLM calibration in static settings. Guo et al. [2017] established Expected Calibration Error (ECE) as the standard metric for probabilistic forecast reliability. Gneiting and Raftery [2007] formalized proper scoring rules—including the Brier score—as the normative framework for evaluating probabilistic predictions. Recent work has documented systematic overconfidence in LLMs across factual question-answering and reasoning tasks [Kadavath et al., 2022, Xiong et al., 2024]. Our contribution extends calibration analysis to a setting where the uncertainty object is a *latent behavioral policy* rather than a factual claim, and where the calibration signal is tied to one-step-ahead opponent prediction rather than retrieved knowledge.

2.4 Faithful Reasoning and Belief–Action Decoupling

The relationship between verbalized reasoning and underlying computation is deeply contested. Turpin et al. [2023] demonstrated that chain-of-thought explanations can systematically misrepresent the true reasons for a model’s prediction—CoT outputs are plausible yet unfaithful when biasing features are added to model inputs. This work motivates our central concern: if reasoning traces can be post-hoc rationalizations rather than drivers of action, the same decoupling may apply to belief posteriors. Wei et al. [2022] showed that chain-of-thought prompting significantly improves reasoning task performance, while Wang et al. [2023] further improved robustness via self-consistency decoding. These techniques evaluate the *quality* of generated reasoning but not whether the reasoning causally influences behavior. Our Experiment 3 (two-stage B2 protocol) directly tests whether making beliefs maximally salient before action selection resolves the decoupling.

2.5 Opponent Modeling and Theory of Mind

The problem of inferring latent opponent policies maps onto the theory of mind (ToM) literature in cognitive science. Premack and Woodruff [1978] established ToM as the capacity to attribute mental states to others; Wimmer and Perner [1983] operationalized this via the false-belief task. In computational terms, Baker et al. [2011] proposed Bayesian ToM as the normative framework for inferring others’

goals and beliefs from observed behavior—the direct ancestor of our oracle posterior. Recent LLM ToM evaluations have found that large models pass first-order false-belief tests while failing higher-order recursive variants [Kosinski, 2024, Chen et al., 2024]. Critically, none of these works evaluate whether ToM-like inference *drives actions*—they measure inference accuracy, not the inference-to-action coupling. Our H6 metric (deceptive-type accuracy gap) operationalizes ToM depth as the capacity to identify strategically deceptive opponent behavior, which requires second-order attribution: the agent must infer that early cooperation is an instrumental strategy, not a policy signal.

2.6 Causal Evaluation of Reasoning

Causal methods for evaluating LLM reasoning have emerged as a corrective to purely behavioral assessments. Meng et al. [2022] introduced causal tracing to identify factual storage sites in transformer weights. Geiger et al. [2024] developed causal abstraction for testing whether model-internal representations implement specific algorithms. At the behavioral level, counterfactual evaluation—presenting modified inputs and measuring output changes—has been applied to logical consistency [Longpre et al., 2021] and compositional generalization [Lake and Baroni, 2018]. Our Posterior Intervention experiment (Stage 4) is the behavioral-level analog: we construct counterfactual posteriors and test whether model actions change in the direction prescribed by expected utility, enabling a causal test of the belief–action link without access to model internals.

2.7 Positioning and Gap

Table 1 summarizes the dimensions covered by the most closely related work. No prior benchmark jointly addresses all four components of the belief–action pipeline:

$$\underbrace{\text{history}}_{\text{observation}} \longrightarrow \underbrace{\hat{P}(\theta \mid h_t)}_{\text{elicited belief}} \xrightarrow{?} \underbrace{a^{\text{EU}}(\hat{P})}_{\text{normative action}} \longrightarrow \underbrace{a_t}_{\text{realized action}} \quad (1)$$

where the central arrow (?) is the causal link our benchmark is designed to test. Existing work measures outcomes at the first and last nodes; we instrument the intermediate nodes and test the arrows.

Table 1: Comparison of related benchmarks across evaluation dimensions. ✓ = explicitly evaluated; ◦ = partially; — = not addressed.

Work	Repeated games	Belief elicitation	Calibration	Coupling test	Causal intervention
Akata et al. [2025]	✓	—	—	—	—
Duan et al. [2024]	✓	—	—	—	—
Phelps and Russell [2025]	✓	—	—	—	—
Binz and Schulz [2023b]	—	—	◦	—	—
Guo et al. [2017]	—	—	✓	—	—
Turpin et al. [2023]	—	—	—	◦	✓
Horton et al. [2023]	◦	—	—	—	—
Ours	✓	✓	✓	✓	✓

3 Problem Setup and Notation

3.1 Environment and Interaction Protocol

We consider a repeated two-player game under partial observability. At each round $t \in \{1, \dots, T\}$ with $T = 10$, an LLM agent selects an action $a_t \in \mathcal{A} = \{C, D\}$ (Cooperate or Defect) simultaneously with

an opponent whose action $a_t^{\text{opp}} \in \mathcal{A}$ is determined by a fixed hidden policy. Both players receive payoffs according to a game-specific utility function:

$$r_t = u(a_t, a_t^{\text{opp}}), \quad u : \mathcal{A} \times \mathcal{A} \rightarrow \mathbb{R}, \quad (2)$$

where u is defined by one of four payoff matrices (Section 4, Table 2); payoffs are always reported as the agent’s return, i.e. index 0 of the tuple $(u_{\text{self}}, u_{\text{opp}})$.

The agent observes the full interaction history at each round:

$$h_t = ((a_1, a_1^{\text{opp}}, r_1), \dots, (a_{t-1}, a_{t-1}^{\text{opp}}, r_{t-1})), \quad (3)$$

with $h_1 = \emptyset$. The opponent never observes the agent’s internal state; the only shared signal is the action pair each round.

The agent’s objective is to maximize cumulative reward over the episode:

$$\max_{a_1, \dots, a_T} \sum_{t=1}^T r_t. \quad (4)$$

3.2 Hidden Opponent Type Model

The opponent’s behavior is governed by a latent type $\theta \in \Theta$, drawn uniformly at episode start and held fixed for all T rounds:

$$\theta \sim \text{Uniform}(\Theta), \quad |\Theta| = 9. \quad (5)$$

The opponent acts according to a policy $\pi_\theta : \mathcal{H} \rightarrow \Delta(\mathcal{A})$, which maps interaction histories to action distributions. Crucially, π_θ is defined *from the opponent’s perspective*: the history tuple is (a_t^{opp}, a_t) —opponent’s own action first, agent’s action second—matching the convention in `opponents.py`. The oracle likelihood computation builds this perspective-flipped history incrementally as (a_t^{opp}, a_t) at each round.

The nine opponent types $\Theta = \{\theta_i\}_{i=1}^9$ span three behavioral classes:

Deterministic stationary. `always_cooperate` ($\pi_\theta \equiv C$) and `always_defect` ($\pi_\theta \equiv D$) are trivially identifiable from round 1.

Deterministic history-dependent. `tit_for_tat`: $a_t^{\text{opp}} = C$ if $t = 1$, else a_{t-1} (agent’s last action). `grim_trigger`: C until the agent defects once, then D forever. `win_stay_lose_shift`: repeat previous action if $r_{t-1}^{\text{opp}} \geq (r_{\max} + r_{\min})/2$, else switch; the threshold is game-specific. `gradual_defector`: cooperates initially; after the agent’s k -th cumulative defection, retaliates with exactly k consecutive defections before returning to cooperation. `adaptive_mirror`: cooperates iff the agent’s cooperation rate over the last $w = 5$ rounds satisfies $\text{CR}_{[-5:]} > 0.6$.

Stochastic. `noisy_tit_for_tat`: tit-for-tat base action flipped independently with probability $\varepsilon_{\text{flip}} = 0.1$ each round. `deceptive_opportunist`: cooperates deterministically for rounds 1–3 ($P(C) \approx 1 - \varepsilon_{\text{lik}}$), then defects with probability 0.80 and cooperates with probability 0.20 for all subsequent rounds.

3.3 Belief Representation

The agent is assumed to maintain a posterior belief over opponent types given history:

$$b_t = \{b_t(\theta)\}_{\theta \in \Theta}, \quad b_t(\theta) = P(\theta \mid h_t), \quad \sum_{\theta \in \Theta} b_t(\theta) = 1. \quad (6)$$

LLMs do not expose internal beliefs directly. We therefore elicit b_t through structured prompt outputs at probe rounds $t \in \{3, 6, 10\}$, requesting a probability simplex over all $|\Theta| = 9$ types as a required JSON field. The elicited distribution $\hat{P}(\theta | h_t)$ is treated as the model’s externalized belief and is the primary object of evaluation in Experiments 2, 3, and 6.

3.4 Oracle Bayesian Posterior

We define a normative reference posterior $P^*(\theta | h_t)$ computed offline by the evaluator. Starting from the uniform prior $P(\theta) = 1/|\Theta|$, the posterior is computed in log-space for numerical stability:

$$\log P^*(\theta | h_t) = \log P(\theta) + \sum_{\tau=1}^{t-1} \log \max(P_\theta(a_\tau^{\text{opp}} | h_\tau), 10^{-300}), \quad (7)$$

where the floor 10^{-300} prevents numerical underflow before taking the logarithm. Normalization uses the log-sum-exp identity:

$$\log Z = \max_{\theta} \log P^*(\theta | h_t) + \log \sum_{\theta'} \exp(\log P^*(\theta' | h_t) - \max_{\theta''} \log P^*(\theta'' | h_t)). \quad (8)$$

After exponentiation and normalization by Z , an additive smoothing step is applied:

$$P_\varepsilon^*(\theta | h_t) = (1 - \varepsilon_s) \tilde{P}^*(\theta | h_t) + \frac{\varepsilon_s}{|\Theta|}, \quad \varepsilon_s = 10^{-6}, \quad (9)$$

followed by renormalization to unity. Throughout this paper, $P^*(\theta | h_t)$ refers to the final smoothed and renormalized posterior. The final renormalization is a numerical precision step, as the smoothed distribution already sums to unity by construction

Likelihood model. The per-round likelihood $P_\theta(a_\tau^{\text{opp}} | h_\tau)$ is defined as:

$$P_\theta(a | h) = \begin{cases} 1 - \varepsilon_{\text{lik}} & \text{deterministic } \theta, a = \hat{a}_\theta(h) \\ \varepsilon_{\text{lik}} & \text{deterministic } \theta, a \neq \hat{a}_\theta(h) \\ 0.9 & \theta = \text{noisy_tit_for_tat}, a = \text{base}(h) \\ 0.1 & \theta = \text{noisy_tit_for_tat}, a \neq \text{base}(h) \\ 1 - \varepsilon_{\text{lik}} & \theta = \text{deceptive_opportunist}, |h| < 3, a = C \\ 0.8 & \theta = \text{deceptive_opportunist}, |h| \geq 3, a = D \\ 0.2 & \theta = \text{deceptive_opportunist}, |h| \geq 3, a = C \end{cases} \quad (10)$$

where $\varepsilon_{\text{lik}} = 0.01$, $\hat{a}_\theta(h)$ is the deterministic action type θ would take given history h , and $\text{base}(h)$ for `noisy_tit_for_tat` is C if $h = \emptyset$, else the agent’s last action $a_{|h|}$ (index 1 of the last opponent-perspective tuple).

3.5 Decision-Theoretic Action Selection

Given a belief distribution b_t over Θ , the normatively rational action is the one maximizing expected immediate payoff:

$$a^{\text{EU}}(b_t) = \arg \max_{a \in \mathcal{A}} \sum_{\theta \in \Theta} b_t(\theta) u(a, \hat{a}_\theta(h_t)), \quad (11)$$

where $\hat{a}_\theta(h_t)$ is the action type θ would take given history h_t , computed via the deterministic or stochastic policy π_θ (using the fixed episode seed s for any stochastic draws). Ties at $\text{EU}(C | b_t) = \text{EU}(D | b_t)$ are broken in favor of C .

We additionally define the *EU margin* as a continuous decision-surface signal:

$$\Delta_{\text{EU}}(b_t) = \sum_{\theta} b_t(\theta) u(D, \hat{a}_{\theta}) - \sum_{\theta} b_t(\theta) u(C, \hat{a}_{\theta}), \quad (12)$$

with the convention that positive values indicate D is EU-optimal and negative values indicate C . This continuous measure complements the binary a^{EU} in cases where the optimal action does not change under posterior perturbation but the decision surface shifts meaningfully—critical for dominant-strategy games such as the Prisoner’s Dilemma (Section 4.7).

3.6 Identifiability

A history h_t is *identifiable* if the oracle posterior concentrates sufficient probability mass on a single type:

$$\text{Identifiable}(h_t) = \mathbf{1} \left[\max_{\theta \in \Theta} P^*(\theta \mid h_t) \geq \tau \right], \quad \tau = 0.60. \quad (13)$$

Identifiable states are those where the available evidence suffices to reliably attribute behavior to a specific type; non-identifiable states reflect genuine ambiguity in the oracle’s inference. Type-accuracy metrics (Section 5) are reported both overall and conditioned on identifiable states to prevent inflating scores at evidence-poor early rounds. Sensitivity analyses are conducted at $\tau \in \{0.40, 0.60, 0.80\}$.

4 The Benchmark Framework

We introduce a benchmark for evaluating belief dynamics and belief-conditioned action in LLMs through controlled repeated strategic interaction. The framework is organized into five pipeline stages, all operating over a single shared interaction trace per condition, thus avoiding redundant API calls while enabling multi-faceted evaluation. We define a *condition* as the tuple $(\mathcal{G}, \theta, i, s)$ where $\mathcal{G}$ is the game, $\theta \in \Theta$ is the opponent type, i is the episode index, and s is a deterministic seed computed as:

$$s = s_0 + \text{hash}(\mathcal{G}, \theta, i) \bmod 2^{31} \quad (14)$$

with base seed $s_0 = 42$. This ensures full reproducibility across providers and models while guaranteeing identical opponent behavioral sequences across all five stages within the same condition.

4.1 Environment: Repeated 2×2 Games with Hidden Types

At each round $t \in \{1, \dots, T\}$ with $T = 10$, the LLM agent selects an action $a_t \in \{C, D\}$ (Cooperate or Defect) simultaneously with an opponent whose action a_t^{opp} is generated by a fixed, hidden policy type θ drawn once at episode start. The agent receives payoff $r_t = R(a_t, a_t^{\text{opp}})$ and observes the full interaction history:

$$h_t = ((a_1, a_1^{\text{opp}}, r_1), \dots, (a_{t-1}, a_{t-1}^{\text{opp}}, r_{t-1})). \quad (15)$$

The type θ is never revealed to the model; only the evaluator has access to it. This constitutes a partially observable setting in which rational action requires forming and updating a belief b_t over the latent type space Θ .

Game structures. We employ four canonical 2×2 games spanning qualitatively distinct incentive regimes. Their payoff matrices (agent, opponent) are given in Table 2.

The selection of these four games is not cosmetic. The Prisoner’s Dilemma provides a dominant-strategy baseline where defection is always Nash-optimal regardless of opponent type, allowing us to test whether belief updates propagate into actions in the absence of strategic necessity. Stag Hunt introduces a coordination equilibrium where cooperation is *only* rational if the agent assigns sufficient

Table 2: Payoff matrices for the four games. Each cell gives (agent payoff, opponent payoff) for outcome (row action, column action). C = Cooperate, D = Defect.

Game	(C,C)	(C,D)	(D,C)	(D,D)
Prisoner’s Dilemma	(3, 3)	(0, 5)	(5, 0)	(1, 1)
Stag Hunt	(4, 4)	(0, 3)	(3, 0)	(2, 2)
Chicken	(3, 3)	(1, 5)	(5, 1)	(0, 0)
Battle of the Sexes	(3, 2)	(0, 0)	(0, 0)	(2, 3)

posterior mass to reciprocating types, directly coupling belief quality to action quality. Chicken inverts the risk structure—mutual defection is the worst outcome—so misidentification of opponent type has asymmetric consequences. Battle of the Sexes features two pure Nash equilibria with opposite payoff rankings, testing whether beliefs over coordination-seeking versus anti-coordination types translate into equilibrium selection. Together, these games ensure that different incentive structures invert the EU-optimal action under different posteriors, isolating whether action selection is belief-driven or game-structure heuristic [Camerer, 2003, Osborne and Rubinstein, 1994].

Opponent type taxonomy. The hidden type θ is drawn uniformly from a set of nine deterministic and stochastic policies Θ , listed in Table 3.

Table 3: Opponent policy types in Θ . Stochastic types have explicit mixing probabilities; all others are deterministic given history.

Type	Behavioral Rule	Stochastic?
<code>always_cooperate</code>	Always plays C	No
<code>always_defect</code>	Always plays D	No
<code>tit_for_tat</code>	C on round 1; copies agent’s last action	No
<code>grim_trigger</code>	C until first agent D , then D forever	No
<code>win_stay_lose_shift</code>	Repeats action if payoff $\geq$ midpoint; else switches	No
<code>noisy_tit_for_tat</code>	Tit-for-tat with flip probability $\epsilon = 0.10$	Yes
<code>deceptive_opportunist</code>	C for rounds 1–3; then D w.p. 0.80, C w.p. 0.20	Yes
<code>gradual_defector</code>	Retaliates with k consecutive D s per k -th agent defection	No
<code>adaptive_mirror</code>	C iff agent cooperation rate in last 5 rounds > 0.60	No

We classify these types along two axes: *behavioral predictability* (deterministic vs. stochastic) and *strategic intent* (cooperative, adversarial, retaliatory, and deceptive). The deceptive subset $\Theta_{\text{dec}} = \{\text{deceptive_opportunist}, \text{gradual_defector}\}$ is of particular theoretical interest: these types instrumentally generate early cooperative signals to suppress the agent’s posterior over adversarial types, before shifting to exploitation. Identifying them correctly requires second-order inference—the agent must infer that early cooperation is a strategic instrument, not a policy signal [Premack and Woodruff, 1978, Baker et al., 2011].

4.2 Oracle Bayesian Posterior

A normative reference for belief quality is provided by the oracle Bayesian posterior $P^*(\theta \mid h_t)$, computed offline by the evaluator over the known type set and observed history. Starting from a uniform prior

$P(\theta) = 1/|\Theta|$, the oracle updates via Bayes’ rule:

$$\log P^*(\theta \mid h_t) = \log P(\theta) + \sum_{\tau=1}^{t-1} \log P_\theta(a_\tau^{\text{opp}} \mid h_\tau), \quad (16)$$

computed in log-space for numerical stability and normalized via the log-sum-exp identity. The likelihood model $P_\theta(a \mid h)$ is defined as:

$$P_\theta(a \mid h) = \begin{cases} 1 - \varepsilon_{\text{lik}} & \text{if } a = \hat{a}_\theta(h) \text{ (deterministic match)} \\ \varepsilon_{\text{lik}} & \text{if } a \neq \hat{a}_\theta(h) \text{ (deterministic mismatch)} \\ 0.9 & \text{if } \theta = \text{noisy_tit_for_tat}, a = \text{base action} \\ 0.8 & \text{if } \theta = \text{deceptive_opportunist}, a = D, t > 3 \end{cases} \quad (17)$$

where $\hat{a}_\theta(h)$ is the action that type θ would deterministically take given history h , and $\varepsilon_{\text{lik}} = 0.01$ is the noise floor preventing zero likelihoods. After normalization, an additive ε -smoothing step is applied:

$$P_\varepsilon^*(\theta \mid h_t) = (1 - \varepsilon_s) P^*(\theta \mid h_t) + \frac{\varepsilon_s}{|\Theta|}, \quad \varepsilon_s = 10^{-6}, \quad (18)$$

followed by renormalization. This two-layer approach—likelihood noise floor ε_{lik} and posterior smoothing ε_s —prevents degenerate collapse to zero-probability assignments across the full episode horizon.

Identifiability. A history h_t is deemed *identifiable* if the oracle posterior places sufficient mass on a single type:

$$\text{Identifiable}(h_t) = \mathbf{1} \left[\max_{\theta \in \Theta} P^*(\theta \mid h_t) \geq \tau \right], \quad \tau = 0.60. \quad (19)$$

We report sensitivity analyses at $\tau \in \{0.40, 0.60, 0.80\}$. Only identifiable states are used as the denominator in type-accuracy metrics for Experiment 2, preventing inflation of accuracy scores at early, evidence-poor rounds.

4.3 Pipeline Architecture

The full evaluation pipeline runs all five experimental stages over a single shared condition, using the same opponent seed s (Equation 14) to ensure identical opponent sequences across stages. This design eliminates the confound of different behavioral sequences when comparing, for example, baseline payoff (Stage 1) with oracle-injected payoff (Stage 3). Table 4 summarizes the stages, their API call counts, and the experiments they feed.

Table 4: Pipeline stage summary. Call counts are per episode per condition. Stage 5 is entirely offline (zero API calls).

Stage	Description	API Calls	Experiments
1	Base gameplay + belief probes	13	Exp. 1, 2, 3-A, 3-B
2	Two-stage belief-action probes	6	Exp. 3-B2
3	Oracle-injected gameplay	10	Exp. 3-C
4	Posterior intervention	0–12	Exp. 4
5	Cognitive theory metrics (offline)	0	Exp. 6

All calls use temperature = 0.0 and top- $p = 1.0$ for determinism, with maximum token budgets of 512 (action calls) and 1024 (probe calls). Responses are constrained to strict JSON schemas (action-only, probe, or belief-only schemas) via structured output mode, eliminating free-text parsing variance. The large system prompt (≈ 350 tokens) is shared and cached across all rounds within an episode, keeping per-round cost proportional to the ≈ 14 -token history increment per round.

4.4 Stage 1: Baseline Gameplay and Belief Elicitation

Stage 1 runs a complete $T = 10$ round episode. At each non-probe round $t \notin \{3, 6, 10\}$, a single API call with the action-only schema produces $a_t \in \{C, D\}$. At designated probe rounds $t \in \{3, 6, 10\}$, an *additional, decoupled* probe call is made over the same history, requesting the extended schema with four required fields:

- `action`: the model’s move for round t ,
- `predicted_next_action`: one-step-ahead opponent prediction,
- `confidence`: scalar confidence in the one-step prediction ($\in [0, 1]$),
- `posterior_over_types`: a probability simplex over all $|\Theta| = 9$ types, constrained to sum to 1.0.

The decoupling of the probe call from gameplay is a deliberate design choice: the model plays its action via the action call first, and the probe elicits beliefs as a parallel query over the same history. This prevents the cognitive demand of belief elicitation from modifying the action selection that constitutes the behavioral baseline for Experiment 1. The probe call does *not* advance the game state; the opponent still responds to the action call’s output. Stage 1 thus generates 13 API calls per episode: 10 action calls and 3 probe calls.

The raw model posterior $\hat{P}$ from each probe is normalized to the probability simplex by clipping negative values to 0 and dividing by the sum; if the sum is zero, a uniform prior over $|\Theta|$ types is substituted. The oracle posterior $P^*(\theta | h_t)$ (Equation 16) is computed at each probe round and stored alongside the model posterior for offline scoring.

4.5 Stage 2: Two-Stage Belief-to-Action Protocol (B2)

Stage 2 isolates the coupling between explicit belief and action by implementing a two-call sequence at each of the three probe rounds ($t \in \{3, 6, 10\}$), yielding 6 additional calls per episode.

Call 1 (Belief-only) queries the model with the same history as Stage 1 but requests only the belief-schema fields (`posterior_over_types`, `confidence`, `predicted_next_action`) without the `action` field. This forces belief articulation prior to any action consideration.

Call 2 (Action-from-posterior) presents the model with the same history *plus* its own elicited posterior from Call 1, rendered as a top-5 probability table, and asks for a single action under the instruction to maximize expected payoff given those beliefs.

This design tests the *deliberation hypothesis*: that making beliefs maximally salient and structurally prior to action selection improves EU-consistency. If coupling failures persist under B2, they cannot be attributed to prompt-format artifacts (belief and action requested in the same call), pointing instead to deeper representational decoupling [Evans, 2003, Kahneman, 2011].

4.6 Stage 3: Oracle-Injected Gameplay

Stage 3 provides an upper bound on performance achievable when belief formation is bypassed. At each round t , the ground-truth oracle posterior $P^*(\theta | h_t)$ (Equation 16) is included in the user prompt as a top-5 probability table, and the model is instructed to select the action maximizing expected immediate payoff under the provided distribution. The game runs for the full $T = 10$ rounds with the same opponent seed as Stage 1, yielding 10 API calls per episode.

The payoff improvement:

$$\Delta_{\text{BUC}} = \bar{R}_{\text{oracle}}^{\text{norm}} - \bar{R}_{\text{baseline}}^{\text{norm}} \quad (20)$$

quantifies *Belief Use Capability*: the value of accurate beliefs that the model fails to realize through its own inference. A model with $\Delta_{\text{BUC}} \approx 0$ infers beliefs of near-oracle quality; a model with $\Delta_{\text{BUC}} \gg 0$ can exploit accurate beliefs when provided but fails to form them independently.

4.7 Stage 4: Posterior Intervention

Stage 4 implements a causal manipulation of the belief distribution. For each probe round $t \in \{3, 6, 10\}$, we construct two counterfactual posteriors from the oracle $P^*(\theta \mid h_t)$.

Adversarial Flip. All probability mass is concentrated on the oracle’s *least likely* type $\theta_{\min} = \arg \min_{\theta} P^*(\theta \mid h_t)$:

$$P_{\text{adv}}^{\text{cf}}(\theta) = \begin{cases} 1 - \eta & \theta = \theta_{\min} \\ \frac{\eta}{|\Theta| - 1} & \theta \neq \theta_{\min} \end{cases} \quad \eta = 0.01. \quad (21)$$

This creates the maximal possible posterior perturbation—a belief diametrically opposed to the oracle.

Plausible Perturbation. The oracle posterior is attenuated by mixing with the uniform distribution:

$$P_{\text{pp}}^{\text{cf}}(\theta) \propto (1 - \lambda) P^*(\theta \mid h_t) + \frac{\lambda}{|\Theta|}, \quad \lambda = 0.80, \quad (22)$$

followed by renormalization. With $\lambda = 0.80$, this provides a strong but directionally coherent attenuation of the oracle signal.

Validity filter. A probe state is evaluated only if the posterior transformation is *informationally meaningful*. We define a state as valid if at least one of the following conditions holds:

$$\text{(EU-flip): } a^{\text{EU}}(P^*) \neq a^{\text{EU}}(P^{\text{cf}}) \quad (23)$$

$$\text{(Margin shift): } |\Delta_{\text{EU}}(P^{\text{cf}}) - \Delta_{\text{EU}}(P^*)| \geq \alpha (r_{\max} - r_{\min}), \quad (24)$$

where $\Delta_{\text{EU}}(P) = \mathbb{E}_{\theta \sim P}[r(D, \hat{a}_{\theta})] - \mathbb{E}_{\theta \sim P}[r(C, \hat{a}_{\theta})]$ is the EU margin, and $\alpha = 0.05$ is the game-scaled threshold with sensitivity analyses at $\alpha \in \{0.03, 0.05, 0.10\}$.

The margin criterion (Equation 24) is critical for dominant-strategy games such as the Prisoner’s Dilemma, where $a^{\text{EU}}(P) = D$ for *any* posterior P , making the EU-flip condition (23) impossible to satisfy. In such games, the margin still shifts meaningfully under posterior perturbation, and excluding these states would systematically bias the valid-probe yield against a game that constitutes one quarter of the evaluation surface.

Valid probe states receive two sequential API calls in randomized order: one with P^* (control) and one with P^{cf} (counterfactual), with no history context. Call order is randomized per probe state using seed $s + t + \text{hash}(\text{transform_name})$ to prevent position bias. Invalid states are recorded with their validity classification (`flip_only`, `margin_only`, `both`, or `invalid`) and excluded from directional accuracy and sensitivity computations.

4.8 Stage 5: Cognitive Theory Metrics (Offline)

Stage 5 performs no additional API calls. It aggregates probe data from Stage 1 (specifically the three probe records per episode at $t \in \{3, 6, 10\}$) and computes six theory-linked diagnostic metrics. Metrics H1, H4, and H7 are computed within the pipeline stage directly over per-episode probe traces; H2, H5, and H6 are derived by the offline analyzer over aggregated episode-level data across all conditions. These are described in detail in Section 5 (Metric Suite); we summarise their theoretical grounding here.

H1 (Bayesian Belief Updating). We compute JS divergence $JS(P^* \parallel \hat{P})$ between the oracle and model posteriors at each probe round. A model performing approximate Bayesian inference should show monotonically decreasing divergence as evidence accumulates [Diaconis and Ylvisaker, 1979].

H2 (EU-Consistent Coherence). The fraction of probe-round actions consistent with the EU-optimal action under the model’s own stated posterior, defined formally in Section 5. This tests whether the belief-to-action mapping is normatively sound [Savage, 1954].

H4 (Metacognitive Calibration). Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) over confidence-correctness pairs ($\text{confidence}_t, \mathbf{1}[\text{predicted_next_action}_t = a_t^{\text{opp}}]$), testing whether the model’s expressed uncertainty aligns with empirical prediction accuracy [Guo et al., 2017].

H5 (Cross-Game Rigidity). The Rigidity Index measures whether the model applies a game-invariant cooperation policy rather than adapting to payoff structure [Simon, 1955].

H6 (Theory-of-Mind Depth). Type-accuracy gap between non-deceptive types and $\Theta_{\text{dec}} = \{\text{deceptive_opportunist}, \text{gradual_defector}\}$, operationalizing ToM depth as the capacity to attribute strategic intent beyond surface behavioral signals [Premack and Woodruff, 1978, Wimmer and Perner, 1983].

H7 (Predictive Processing). Pearson correlation between per-probe surprisal $\delta_t = -\log P_{\text{model}}(a_{t+1}^{\text{opp}} \mid h_t)$ and belief update magnitude $\|\hat{P}_{t+1} - \hat{P}_t\|_1$, testing whether larger prediction errors produce proportionally larger posterior revisions [Friston, 2010, Clark, 2013].

4.9 Experimental Conditions and Sample Size

The full benchmark is a balanced factorial design: 4 games $\times$ 9 opponent types $\times$ 10 episodes = 360 episodes per model. Probe rounds generate 3 observations per episode, yielding 1,080 belief-quality datapoints. Stage 4 generates up to 12 intervention calls per episode (3 probe rounds $\times$ 2 transforms $\times$ 2 posterior queries), for a maximum of 4,320 intervention queries before validity filtering.

Sample size adequacy is evaluated against an estimation precision target of 95% CI half-width ≤ 0.05 on normalized payoff. Under the conservative assumption $\sigma \approx 0.20$ for bounded $[0, 1]$ outcomes:

$$N \geq \left(\frac{1.96 \sigma}{\varepsilon} \right)^2 \approx 62 \text{ episodes per game,} \quad (25)$$

which our design satisfies at 90 episodes per game (10 episodes $\times$ 9 opponent types). All confidence intervals are computed via the normal approximation $\bar{x} \pm 1.96 \cdot \hat{\sigma} / \sqrt{n}$ at the episode level, where $\hat{\sigma}$ is the sample standard deviation with $\text{ddof} = 1$, to account for within-episode dependence [Efron and Tibshirani, 1993].

5 Metric Suite

We define a preregistered set of metrics organized around the four components of the belief–action pipeline: *behavioral performance* (Experiment 1), *belief quality and calibration* (Experiment 2), *belief–action coupling* (Experiment 3), *causal posterior sensitivity* (Experiment 4), and *cognitive theory diagnostics* (Experiment 6). Every metric is computed from data already logged during gameplay; no post-hoc labeling or outcome-dependent filters are introduced. Table 5 provides a consolidated reference.

A key design principle is the *hierarchy of coupling strength* []: metrics at higher levels of the hierarchy make strictly stronger claims about the causal relationship between belief and action. Level 1 (sensitivity) asks whether anything changes; Level 2 (conditional agreement) asks whether changes are directionally consistent with stated beliefs; Level 3 (EU-coherence) asks whether changes are normatively optimal under those beliefs; Level 4 (directional accuracy under intervention) asks whether optimal changes

survive a fully causal posterior manipulation. A model can pass lower levels while failing higher ones, and this stratification is the primary diagnostic instrument of the benchmark.

5.1 Experiment 1: Behavioral Performance Metrics

These metrics establish the raw strategic baseline. They require no belief elicitation and serve as the anchor against which belief-conditioned performance gains are measured in Experiment 3.

Normalized episode payoff. Raw payoffs are incomparable across games with different payoff scales. We therefore normalize per-round payoffs to the $[0, 1]$ interval using the game-specific minimum and maximum:

$$\bar{R}^{\text{norm}} = \frac{1}{T} \sum_{t=1}^T \frac{r_t - r_{\min}^{(\mathcal{G})}}{r_{\max}^{(\mathcal{G})} - r_{\min}^{(\mathcal{G})}}, \quad (26)$$

where $r_{\min}^{(\mathcal{G})}$ and $r_{\max}^{(\mathcal{G})}$ are the minimum and maximum single-round payoffs achievable by the agent in game $\mathcal{G}$, taken from the payoff matrix directly (e.g. $r_{\min}^{\text{PD}} = 0$, $r_{\max}^{\text{PD}} = 5$). Range: $[0, 1]$. This is the primary outcome metric for Experiment 1 and the baseline term in all Δ payoff comparisons in Experiment 3.

Cooperation rate.

$$\text{CR} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[a_t = C]. \quad (27)$$

Range: $[0, 1]$. A secondary behavioral metric used in Experiment 6 to compute the Rigidity Index. Cooperation rate is informative precisely because it should *vary* across games if the model is adapting to incentive structure; a model playing the same CR across all four games is exhibiting heuristic rather than belief-driven behavior.

Opponent-conditional payoff profile. We compute $\bar{R}^{\text{norm}}(\mathcal{G}, \theta)$ for each game–opponent-type cell. This disaggregation allows identification of specific failure modes: high average payoff with low payoff against deceptive opponents signals exploitation vulnerability, not general strategic competence.

5.2 Experiment 2: Belief Quality Metrics

These metrics evaluate how well the model’s stated posterior $\hat{P}(\theta | h_t)$ approximates the oracle posterior $P^*(\theta | h_t)$ at probe rounds $t \in \{3, 6, 10\}$. They measure the *inference and representation* components of the pipeline without any assumption about whether beliefs drive action.

Algorithm 1 describes the full per-episode computation.

KL divergence from oracle (primary).

$$D_{\text{KL}}(P^* \| \hat{P}) = \sum_{\theta \in \Theta} P^*(\theta) \log \frac{P^*(\theta)}{\hat{P}_\varepsilon(\theta)}, \quad (28)$$

where the smoothed model posterior is $\hat{P}_\varepsilon(\theta) = (1 - \varepsilon) \hat{P}(\theta) + \varepsilon / |\Theta|$ with $\varepsilon = 10^{-6}$. D_{KL} is asymmetric and penalizes cases where P^* assigns positive probability to a type that $\hat{P}$ underweights, making it sensitive to missed hypotheses. Range: $[0, \infty)$. We report the mean across probe rounds within an episode.

Algorithm 1 Experiment 2: Belief Quality Computation (per episode)

Require: Episode trace with probe records at $t \in \{3, 6, 10\}$, true opponent type θ^*

Ensure: Per-episode metrics: $\overline{D_{\text{KL}}}$, $\overline{\text{BS}}$, ECE, MCE, type accuracy

```
1:  $kl \leftarrow \emptyset$ ;  $bs \leftarrow \emptyset$ ;  $cal \leftarrow \emptyset$ ;  $type\_correct \leftarrow 0$ 
2: for each probe record  $\rho$  at round  $t$  do
3:    $\hat{P} \leftarrow \text{NORMALIZEPOSTERIOR}(\rho.\text{posterior\_over\_types})$ 
4:    $P^* \leftarrow \text{ORACLEPOSTERIOR}(h_t, \mathcal{G}, s)$  ▷ Eq. 16
5:    $kl \leftarrow kl \cup \{D_{\text{KL}}(P^* \parallel \hat{P})\}$ 
6:    $bs \leftarrow bs \cup \{\text{BS}(\hat{P}, \theta^*)\}$ 
7:    $\tilde{\theta} \leftarrow \arg \max_{\theta} \hat{P}(\theta)$ 
8:   if  $\tilde{\theta} = \theta^*$  then
9:      $type\_correct \leftarrow type\_correct + 1$ 
10:  end if
11:   $pred\_acc \leftarrow \mathbf{1}[\rho.\text{predicted\_next\_action} = a_{t+1}^{\text{opp}}]$ 
12:   $cal \leftarrow cal \cup \{(\rho.\text{confidence}, pred\_acc)\}$ 
13: end for
14: return  $\overline{D_{\text{KL}}}(kl)$ ,  $\overline{\text{BS}}(bs)$ ,  $\text{ECE}(cal)$ ,  $\text{MCE}(cal)$ ,  $type\_correct / |kl|$ 
```

Multiclass Brier score.

$$\text{BS}(\hat{P}, \theta^*) = \sum_{\theta \in \Theta} (\hat{P}(\theta) - \mathbf{1}[\theta = \theta^*])^2. \quad (29)$$

Range: $[0, 2]$ for $|\Theta| = 9$ classes; $\text{BS} = 0$ iff $\hat{P}$ places all mass on the true type. Unlike KL, the Brier score is a proper scoring rule that evaluates forecast accuracy against the realized ground truth θ^* , not against the oracle posterior [Gneiting and Raftery, 2007]. It is therefore sensitive to both inference error and oracle posterior concentration: a model can have low D_{KL} but high BS if the oracle itself is diffuse over several compatible types.

Type accuracy.

$$\text{TypeAcc} = \frac{1}{|\mathcal{P}|} \sum_{\rho \in \mathcal{P}} \mathbf{1} \left[\arg \max_{\theta} \hat{P}_{\rho}(\theta) = \theta^* \right], \quad (30)$$

where $\mathcal{P}$ is the set of probe records. This is reported both overall and separately for identifiable states ($\max_{\theta} P^*(\theta \mid h_t) \geq \tau = 0.60$) to avoid rewarding correct guesses at evidence-poor rounds.

Expected Calibration Error. Calibration evaluates whether the model’s expressed confidence in its one-step-ahead opponent prediction aligns with actual prediction accuracy. We partition predictions into $B = 10$ equal-width confidence bins $\{\mathcal{B}_b\}_{b=1}^B$ and compute:

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{N} |\text{acc}(\mathcal{B}_b) - \overline{\text{conf}}(\mathcal{B}_b)|, \quad (31)$$

where $\text{acc}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_i \mathbf{1}[\hat{a}_i^{\text{opp}} = a_i^{\text{opp}}]$ and $\overline{\text{conf}}(\mathcal{B}_b)$ is the mean confidence in bin b [Guo et al., 2017]. Range: $[0, 1]$; $\text{ECE} = 0$ under perfect calibration. The bin assignment uses the ceiling rule: $b = \min(\lceil \text{confidence} \times B \rceil, B - 1)$.

Maximum Calibration Error.

$$\text{MCE} = \max_{b \in \{1, \dots, B\}} |\text{acc}(\mathcal{B}_b) - \overline{\text{conf}}(\mathcal{B}_b)|. \quad (32)$$

Algorithm 2 EU-Optimal Action Under Posterior

Require: Posterior P over Θ , game $\mathcal{G}$, history h_t , seed s

Ensure: $a^{\text{EU}}(P) \in \{C, D\}$

```
1:  $eu_C \leftarrow 0$ ;  $eu_D \leftarrow 0$ 
2: for  $(\theta, p) \in P$  do
3:    $\hat{a}^\theta \leftarrow \text{GETOPPONENTACTION}(\theta, \text{OPPHISTORY}(h_t), \mathcal{G}, s)$ 
4:    $eu_C += p \cdot R(C, \hat{a}^\theta)$ 
5:    $eu_D += p \cdot R(D, \hat{a}^\theta)$ 
6: end for
7: return  $C$  if  $eu_C \geq eu_D$  else  $D$ 
```

MCE identifies the worst-case calibration failure across any confidence stratum. A model may have low ECE through compensation across bins while still exhibiting severe overconfidence at high-confidence predictions; MCE exposes this [Naeini et al., 2015].

5.3 Experiment 3: Belief–Action Coupling Metrics

Experiment 3 is the central coupling analysis. It compares four prompt variants—**A** (action-only baseline), **B** (single-call probe), **B2** (two-stage: belief-first then action), and **C** (oracle-injected)—all run with identical opponent sequences (same seed). Algorithm 2 defines the EU-optimal action computation shared across all coupling metrics.

Formally, the EU-optimal action is:

$$a^{\text{EU}}(P) = \arg \max_{a \in \{C, D\}} \sum_{\theta \in \Theta} P(\theta) R(a, \hat{a}^\theta(h_t)), \quad (33)$$

where $\hat{a}^\theta(h_t)$ is the action type θ would take deterministically given history h_t and seed s . Ties at $eu_C = eu_D$ are broken in favor of C .

Payoff deltas. The payoff difference between variant X and baseline A quantifies the performance value of belief-related intervention:

$$\Delta_{B-A} = \bar{R}_B^{\text{norm}} - \bar{R}_A^{\text{norm}}, \quad (34)$$

$$\Delta_{B2-A} = \bar{R}_{B2}^{\text{norm}} - \bar{R}_A^{\text{norm}}, \quad (35)$$

$$\Delta_{C-A} = \bar{R}_C^{\text{norm}} - \bar{R}_A^{\text{norm}}. \quad (36)$$

Δ_{C-A} is the *Belief Use Capability* (BUC) [1]: the payoff gain attributable to oracle-accurate beliefs. It is the ceiling that Δ_{B-A} and Δ_{B2-A} are bounded by if belief formation were perfect. The ratio $\Delta_{B2-A}/\Delta_{C-A}$ quantifies what fraction of the oracle gain the two-stage B2 protocol recovers.

Conditional action agreement. Action agreement between variants A and B is computed exclusively on *identifiable* probe states ($\max_{\theta} P^*(\theta \mid h_t) \geq \tau$), where the oracle posterior concentrates sufficient mass to reliably identify an opponent type.

$$\text{CAA} = \frac{\sum_i \mathbf{1}[\text{ident}_i] \cdot \mathbf{1}[a_i^A = a_i^B]}{\sum_i \mathbf{1}[\text{ident}_i]}. \quad (37)$$

High CAA with low EU-coherence (below) is the “consistency without rationality” failure mode: the model is self-consistent but its action is not optimal given its stated belief.

Algorithm 3 Experiment 4: Posterior Intervention (per probe state)

Require: Oracle posterior P^* , game $\mathcal{G}$, probe history h_t , seed s , transform $\mathcal{T}$

Ensure: Validity classification, a^{ctl} , a^{cf} , DirAcc, Sens

```
1:  $P^{\text{cf}} \leftarrow \mathcal{T}(P^*)$  ▷ Eqs. 21, 22
2:  $m^{\text{ctl}} \leftarrow \Delta_{\text{EU}}(P^*, h_t)$ ;  $m^{\text{cf}} \leftarrow \Delta_{\text{EU}}(P^{\text{cf}}, h_t)$  ▷ Eq. 39
3:  $eu\_flip \leftarrow [a^{\text{EU}}(P^*) \neq a^{\text{EU}}(P^{\text{cf}})]$ 
4:  $margin\_shift \leftarrow [|m^{\text{cf}} - m^{\text{ctl}}| \geq \alpha(r_{\max} - r_{\min})]$ ;  $\alpha = 0.05$ 
5: if  $\neg eu\_flip$  and  $\neg margin\_shift$  then
6:   record validity  $\leftarrow$  invalid; skip API calls
7:   return
8: end if
9: classify validity  $\in \{\text{flip\_only}, \text{margin\_only}, \text{both}\}$ 
10: Randomize call order via seed' =  $s + t + \text{hash}(\mathcal{T})$ 
11:  $a^{\text{ctl}} \leftarrow \text{QUERYLLM}(P^*)$ ;  $a^{\text{cf}} \leftarrow \text{QUERYLLM}(P^{\text{cf}})$  ▷ posterior-only prompt, no history
12: return  $a^{\text{ctl}}$ ,  $a^{\text{cf}}$ ,  $\mathbf{1}[a^{\text{cf}} = a^{\text{EU}}(P^{\text{cf}})]$ ,  $\mathbf{1}[a^{\text{cf}} \neq a^{\text{ctl}}]$ 
```

EU-consistent coherence (primary coupling metric).

$$\text{Coh}(V) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[a_i^V = a^{\text{EU}}(\hat{P}_i^V)], \quad (38)$$

where $V \in \{B, B2\}$ is the variant, a_i^V is the model's action, and $\hat{P}_i^V$ is the model's own stated posterior, both from the same probe call (or two-stage pair). $\text{Coh}(V) = 1.0$ implies perfect rational coupling: every action the model takes is the action that maximizes its expected payoff under its own expressed beliefs. $\text{Coh}(V) \approx 0.5$ is the chance level for a two-action choice and indicates complete decoupling.

This is the most demanding metric in the suite. It requires simultaneously accurate posterior expression *and* optimal action selection relative to that posterior—and their conjunction, not either alone.

5.4 Experiment 4: Posterior Intervention Metrics

Experiment 4 elevates coupling analysis from correlation to causation. By injecting counterfactual posteriors while holding the game state fixed, we test whether posterior changes produce action changes of the right sign and magnitude. Algorithm 3 describes the per-probe procedure.

EU margin (continuous validity signal).

$$\Delta_{\text{EU}}(P, h_t) = \underbrace{\sum_{\theta} P(\theta) R(D, \hat{a}^{\theta})}_{\text{EU}(D|P)} - \underbrace{\sum_{\theta} P(\theta) R(C, \hat{a}^{\theta})}_{\text{EU}(C|P)}. \quad (39)$$

Positive values indicate D is EU-optimal; negative values indicate C . This continuous signal complements the binary a^{EU} in validity filtering: a Prisoner's Dilemma probe where D is always optimal may never satisfy the EU-flip criterion but still have a meaningful margin shift under adversarial posterior perturbation.

Validity classification. Each probe state falls into one of four validity classes based on Equations 23 and 24:

$$\text{validity} = \begin{cases} \text{both} & eu_flip \wedge margin_shift \\ \text{flip_only} & eu_flip \wedge \neg margin_shift \\ \text{margin_only} & \neg eu_flip \wedge margin_shift \\ \text{invalid} & \neg eu_flip \wedge \neg margin_shift \end{cases} \quad (40)$$

Only `flip_only`, `margin_only`, and `both` states receive API calls. The distribution of validity classes is reported as a diagnostic; a majority of `margin_only` states indicates the analysis is driven by continuous decision-surface shifts rather than action flips.

Directional accuracy.

$$\text{DirAcc} = \frac{1}{N_{\text{valid}}} \sum_{i \in \text{valid}} \mathbf{1}[a_i^{\text{cf}} = a^{\text{EU}}(P_i^{\text{cf}})]. \quad (41)$$

This asks: when the model is given P^{cf} and P^{cf} prescribes action a^* , does the model select a^* ? Range: $[0, 1]$. Random behavior yields 0.5; perfect causal coupling yields 1.0.

Sensitivity to posterior.

$$\text{Sens} = \frac{1}{N_{\text{valid}}} \sum_{i \in \text{valid}} \mathbf{1}[a_i^{\text{cf}} \neq a_i^{\text{ctl}}]. \quad (42)$$

Sensitivity measures any action change regardless of direction. The gap $\text{Sens} - \text{DirAcc} \geq 0$ quantifies *irrational responsiveness*: the fraction of states where the model does react to the posterior change but moves in the wrong direction. A high gap is a stronger indictment than low DirAcc alone, because it rules out a non-responsive null hypothesis and shows the model is actively driven by beliefs—just not rationally.

Sensitivity analyses for both metrics are reported at threshold levels $\alpha \in \{0.03, 0.05, 0.10\}$.

5.5 Experiment 6: Cognitive Theory Metrics

These metrics are computed entirely offline from Stage 1 probe data. They map LLM behavioral traces onto named cognitive frameworks, providing a theory-level interpretation of the quantitative results. No additional API calls are required.

H1 — Bayesian belief updating (JS divergence).

$$\text{JS}(P^* \parallel \hat{P}) = \frac{1}{2} D_{\text{KL}}(P^* \parallel M) + \frac{1}{2} D_{\text{KL}}(\hat{P} \parallel M), \quad M = \frac{1}{2}(P^* + \hat{P}). \quad (43)$$

Range: $[0, \ln 2] \approx [0, 0.693]$. JS is the primary H1 metric because it is symmetric, bounded, and well-defined even when $\hat{P}$ assigns zero to types that P^* weights positively—a limitation of raw KL divergence. We additionally report mean $D_{\text{KL}}(P^* \parallel \hat{P})$ (Equation 28) as a secondary metric for backward compatibility with Experiment 2. A model performing approximate Bayesian inference should exhibit monotonically decreasing JS across probe rounds $t \in \{3, 6, 10\}$ as evidence accumulates [Diaconis and Ylvisaker, 1979].

H2 — EU-consistent coherence. The same $\text{Coh}(B)$ metric from Experiment 3 (Equation 38) is re-evaluated here as a theory-level diagnostic: coherence maps onto the rational agent assumption that beliefs and actions are jointly governed by expected utility maximization [Savage, 1954]. Coherence below 0.5 is a strict falsification of this assumption.

H4 — Metacognitive calibration (ECE, MCE). ECE and MCE from Equations 31–32 are reinterpreted under the metacognition framework [Flavell, 1979, Guo et al., 2017]: an ideally calibrated agent has complete self-knowledge of its own prediction uncertainty. Systematic $\text{ECE} > 0$ indicates that the model’s expressed confidence is not a faithful signal of its actual predictive accuracy—a form of miscalibrated self-model.

H5 — Cross-game rigidity. We measure whether cooperation rates adapt across the four games’ distinct incentive structures using the Rigidity Index:

$$\text{RI} = 1 - \frac{\text{SD}(\{CR_g\}_{g=1}^4)}{0.57735}, \quad (44)$$

where $\text{SD}(\cdot)$ is the population standard deviation ($\text{ddof}=0$) and $0.57735 = \max \text{SD}$ for four values in $[0, 1]$, achieved by the split $\{0, 0, 1, 1\}$. Range: $[0, 1]$ approximately; $\text{RI} = 1$ implies identical cooperation rates across all games (maximum rigidity), $\text{RI} = 0$ implies maximal cross-game adaptation. Bounded rationality theory predicts $\text{RI} > 0$ for agents with limited representational or computational resources applied to game-specific reasoning [Simon, 1955, Gigerenzer and Goldstein, 1996].

H6 — Theory-of-mind depth (deceptive-type gap).

$$\text{Gap}_{\text{ToM}} = \text{TypeAcc}_{\Theta \setminus \Theta_{\text{dec}}} - \text{TypeAcc}_{\Theta_{\text{dec}}}, \quad (45)$$

where $\Theta_{\text{dec}} = \{\text{deceptive_opportunist}, \text{gradual_defector}\}$ and TypeAcc is the argmax-match accuracy (Equation 30). A positive Gap_{ToM} indicates that the model is systematically misled by early cooperative signals from deceptive types, attributing them to cooperative rather than deceptive policies. This operationalizes Theory-of-Mind depth [Premack and Woodruff, 1978, Wimmer and Perner, 1983]: identifying a deceptive type requires attributing the intent to produce false beliefs in the observer, a second-order mental state attribution. Both $\text{TypeAcc}_{\Theta_{\text{dec}}}$ and $\text{TypeAcc}_{\Theta \setminus \Theta_{\text{dec}}}$ are reported with 95% confidence intervals via stratified bootstrap.

H7 — Predictive processing: surprisal–update coupling. For each probe transition from round t to t' (adjacent probe records), we compute:

$$\delta_t = -\log \tilde{p}_t, \quad \tilde{p}_t = \max(\hat{c}_t \cdot \mathbf{1}[\hat{a}_t^{\text{opp}} = a_t^{\text{opp}}] + (1 - \hat{c}_t) \cdot \mathbf{1}[\hat{a}_t^{\text{opp}} \neq a_t^{\text{opp}}], 0.01), \quad (46)$$

$$\Delta_t = \sum_{\theta \in \Theta} |\hat{P}_{t'}(\theta) - \hat{P}_t(\theta)| \quad (\text{L1 belief update magnitude}), \quad (47)$$

where $\hat{c}_t$ is confidence and $\hat{a}_t^{\text{opp}}$ is `predicted_next_action` from the probe at round t . The surprisal δ_t is constructed from the model’s own stated confidence so that it reflects the model’s *self-assessed* prediction error, not an external ground truth—ensuring the metric is non-leaky.

The coupling coefficient is:

$$\rho_{\text{H7}} = \frac{\sum_i (\delta_i - \bar{\delta})(\Delta_i - \bar{\Delta})}{\sqrt{\sum_i (\delta_i - \bar{\delta})^2} \sqrt{\sum_i (\Delta_i - \bar{\Delta})^2}}, \quad (48)$$

computed using the Pearson correlation formula implemented in `pipeline.py: _pearson_corr()`. The alignment vector uses $\min(|\delta\text{-series}| - 1, |\Delta\text{-series}|)$ pairs, consuming surprisal values from round t against update magnitudes from the transition $t \rightarrow t'$ —matching the causal direction (prediction error at t drives update between t and t'). A positive ρ_{H7} is consistent with predictive processing accounts [Friston, 2010, Clark, 2013]: larger prediction errors should trigger larger posterior revisions.

5.6 Metric Summary and Failure Mode Taxonomy

The metrics form a strict hierarchy of claim strength:

1. **Sensitivity** (Sens): does the model respond to belief changes at all?
2. **Consistency** (CAA): do responses align with stated beliefs?
3. **Rationality** (Coh): are actions EU-optimal under stated beliefs?

Table 5: Complete preregistered metric suite. All metrics are computed from logged data; none depend on post-hoc labeling. CI = 95% cluster bootstrap confidence interval at episode level.

Symbol	Metric	Range	Experiment	Failure diagnosed
$\bar{R}^{\text{norm}}$	Norm. episode payoff	$[0, 1]$	Exp1	Poor strategy
CR	Cooperation rate	$[0, 1]$	Exp1, 6	Rigidity
D_{KL}	KL div. from oracle	$[0, \infty)$	Exp2, 6	Inference error
BS	Multiclass Brier score	$[0, 2]$	Exp2	Forecast error
TypeAcc	Argmax type accuracy	$[0, 1]$	Exp2, 6	Identification failure
ECE	Expected calib. error	$[0, 1]$	Exp2, 6	Overconfidence
MCE	Max calib. error	$[0, 1]$	Exp2, 6	Worst-case miscalib.
Δ_{B-A}	Belief-probe payoff gain	$(-1, 1)$	Exp3	Belief underuse
Δ_{B2-A}	Two-stage payoff gain	$(-1, 1)$	Exp3	Coupling failure
Δ_{C-A}	Oracle payoff gain (BUC)	$(-1, 1)$	Exp3	Belief formation cap
CAA	Conditional action agreement	$[0, 1]$	Exp3	Inconsistency
Coh	EU-consistent coherence	$[0, 1]$	Exp3, 6	Irrational decoupling
DirAcc	Directional accuracy	$[0, 1]$	Exp4	Causal failure
Sens	Sensitivity to posterior	$[0, 1]$	Exp4	Non-responsiveness
JS	JS divergence	$[0, \ln 2]$	Exp6	Bayesian update failure
RI	Rigidity index	$\approx [0, 1]$	Exp6	Bounded rationality
Gap _{ToM}	Deceptive-type accuracy gap	$[-1, 1]$	Exp6	ToM depth failure
ρ_{H7}	Surprisal–update correlation	$[-1, 1]$	Exp6	Pred.-processing failure

4. **Causal rationality** (DirAcc): does injected belief produce EU-optimal action under full causal control?

A model can satisfy Level 1 while failing all higher levels (irrational responsiveness), or can satisfy Level 2 while failing Level 3 (self-consistent but suboptimal), or can satisfy Level 3 in correlational play while failing Level 4 under causal intervention (belief–action coherence driven by confounded history rather than genuine posterior use). This layered structure is the core diagnostic instrument of the benchmark; results must be interpreted jointly across all four levels.

6 Experimental Setup

This section describes the models evaluated, the multi-provider execution infrastructure, and the reproducibility mechanisms that collectively guarantee every reported number is traceable to a specific model string, seed, and prompt hash.

6.1 Models Evaluated

We evaluate six model configurations spanning five provider families. Table 6 lists each configuration with its exact model identifier string as used in API calls. These strings are hardcoded as defaults in `config.py` and may be overridden per-run via environment variables, ensuring full reproducibility.

All models are evaluated with temperature = 0.0 and top- p = 1.0, enforcing deterministic decoding wherever the provider API supports it. OpenAI o-series reasoning models (o1, o3, o4 prefixes) do not accept temperature or top- p parameters and are evaluated under their default sampling configuration [OpenAI, 2024, Team et al., 2024, Meta AI, 2024]. Gemini 2.5 Pro requires a minimum thinking budget; we

Table 6: Models evaluated. Provider column indicates the API backend; model strings are the exact identifiers passed to each provider’s inference endpoint. Reasoning models (o-series) are excluded from temperature and top- p constraints due to API restrictions.

Provider Key	Model String	Backend
openai	gpt-5.4-nano	OpenAI Chat Completion
gemini	gemini-2.0-flash	Google GenAI
vertex_gemini	gemini-2.5-pro	Vertex AI (GCP)
vertex_llama	meta/llama-4-maverick-17b-128e-instruct-maas	Vertex AI Model Garden
groq	meta/llama-3.1-8B-instruct	GROQ API
vertex_deepseek	deepseek-ai/deepseek-v3.2-maas	Vertex AI Model Garden

set `thinking_budget=128` tokens and raise the maximum output budget to 4096 tokens to prevent JSON truncation during probe calls.

6.2 JSON Output Enforcement

Structured JSON output is critical for this benchmark: every game action, belief posterior, and confidence score is parsed directly from the model response without free-text post-processing. Provider APIs offer varying degrees of JSON enforcement, and we apply the strongest available mechanism for each:

- **OpenAI:** `response_format = {type: json_schema, json_schema: {strict: true, schema: ...}}`. Strict mode guarantees schema-valid output at the API level, eliminating parse failures for well-formed requests.
- **Gemini (direct and Vertex):** `response_mime_type = "application/json"` with `response_schema` set to the stripped schema (fields `additionalProperties`, `$schema`, and `title` removed as they are unsupported by the GenAI SDK). If a schema field error is raised, the request is retried without the schema but with MIME-type enforcement retained.
- **All other providers (Groq, Vertex Llama/DeepSeek):** `response_format = {type: json_object}`, which enforces JSON syntax without field-level schema validation.

Responses are parsed by first attempting `json.loads()` on the full raw text; on `JSONDecodeError`, a fallback extracts the substring from the first `{` to the last `}` and re-attempts parsing. Failed parses return an empty dict `{}`, which is counted as a parse failure in integrity reporting. Schema failures (valid JSON but missing required fields) are tracked separately.

6.3 Retry and Rate-Limit Protocol

All API calls are wrapped in a uniform retry loop with exponential backoff. The backoff delay after attempt k is:

$$d_k = \min(d_0 \cdot 2^k, d_{\max}), \quad d_0 = 1.0 \text{ s}, \quad d_{\max} = 30.0 \text{ s}, \quad (49)$$

giving the sequence $\{1, 2, 4, 8, 16, 30\}$ seconds over at most $K = 6$ attempts before raising a terminal `RuntimeError`. HTTP 429 (rate limit exceeded) and HTTP 5xx (server error) are retried; 4xx client errors (excluding 429) are raised immediately. Vertex AI Anthropic additionally enforces a minimum inter-call delay of 4.0 seconds due to its low token-per-minute quota ($\approx 15\text{k}$ TPM), applied after each successful response as $\max(0, 4.0 - \text{latency})$ seconds of sleep.

6.4 Prompt Auditing and Integrity Logging

Every API call is assigned a 16-character prompt hash:

$$h_{\text{prompt}} = \text{SHA-256}(\text{system_prompt} \parallel \text{" " } \parallel \text{user_prompt})[:16], \quad (50)$$

computed before the request is sent and stored alongside the response metadata (provider, model string, response ID, latency, token usage, cached token count). This hash enables post-hoc verification that any two calls with identical hashes received identical prompts, and supports detection of accidental prompt mutations during resumption from checkpoint.

Per-call metadata includes: stage (e.g. `s1_action`, `s1_probe`, `s2_belief`, `s3_oracle`, `s4`), episode ID, round index t , and transform name for Stage 4 calls. This granularity allows all parse failures and schema failures to be attributed to a specific stage, game, and round rather than aggregated globally.

6.5 Checkpointing and Resumption

Long evaluation runs across 360 conditions and 11 models are vulnerable to interruption by API errors, rate-limit exhaustion, or infrastructure failures. We implement a condition-granular checkpoint system that persists state after every completed condition:

1. After each condition $(\mathcal{G}, \theta, i, s)$ completes all five stages, the result is appended to the `completed` list.
2. The checkpoint is written atomically: the serialized JSON is first written to a `.tmp` file, then renamed via `os.replace()`, which is atomic on POSIX systems. This prevents corrupt checkpoint files from partially-written states.
3. On restart, the pipeline loads the checkpoint, restores the `completed` and `pending` lists, and resumes from the first entry in `pending` without re-running any completed condition.
4. API log entries (`api_log` per stage) are stripped from checkpoint data before serialization to bound file size; they are retained in memory during a run but not persisted.

Checkpoint files are named `ckpt_pipeline_{provider}_{model}_{tag}.json` and stored under `checkpoints/`. Multiple runs with different `tag` values can coexist without collision, supporting parallel execution across providers.

6.6 Total Computation Budget

Table 7 summarizes the API call budget per model under the full 360-condition design. The figures account for the worst-case Stage 4 call count (all probe states valid, both transforms applied).

Table 7: API call budget per model under the full factorial design (360 conditions $\times$ 10 episodes per game = 360 total conditions). Stage 4 figures assume maximum valid-probe yield. Stage 5 is offline (0 calls). Numbers are per-model totals.

Stage	Calls / cond.	Total calls	Feeds
Stage 1 (gameplay + probes)	13	4,680	Exp. 1, 2, 3-A/B
Stage 2 (B2 two-stage)	6	2,160	Exp. 3-B2
Stage 3 (oracle-injected)	10	3,600	Exp. 3-C
Stage 4 (intervention)	0–12	$\leq 4,320$	Exp. 4
Stage 5 (offline)	0	0	Exp. 6
Total		$\leq 14,760$	

At an average of ~ 400 tokens per action call and ~ 800 tokens per probe call, the total token consumption per model is approximately 6–9 million tokens, dominated by the growing history component (≈ 14 tokens per round) across Stage 1 and Stage 3. Prompt caching reduces effective cost for providers that support it (OpenAI, Gemini): the fixed system prompt (≈ 350 tokens) is cached after the first call in each episode and not rebilled on subsequent rounds, yielding a cache hit on 7 of the 13 Stage 1 calls per episode.

7 Results

We evaluate six models across 360 episodes per model ($4 \text{ games} \times 9 \text{ opponent types} \times 10 \text{ episodes}$), yielding 1,080 probe observations and up to 1,866 valid intervention probes per model. All results are reported as means with 95% confidence intervals computed via the normal approximation $\bar{x} \pm 1.96 \cdot \hat{\sigma} / \sqrt{n}$ at the episode level ($\hat{\sigma}$ uses $\text{ddof} = 1$). Table 8 provides the consolidated cross-model overview.

7.1 Master Comparison Table

Table 8: Cross-model summary of all primary metrics organised by pipeline stage. Arrows indicate the desirable direction. **Bold** = best per metric; underline = worst. [†] Negative BUC means oracle beliefs reduce performance. * LLaMA 3.1 8B’s low ECE is an artifact of near-uniform outputs.

Model	$\bar{R}^{\text{norm}}$ ↑	KL ↓	Brier ↓	TypeAcc ↑	ECE ↓	Coh(B) ↑	BUC ↑	DirAcc ↑	JS ↓	RI ↓
DeepSeek V3.2	0.634	7.75	1.030	0.253	0.211	0.912	+0.030	0.701	0.315	0.420
Gemini 2.5 Pro	0.682	8.11	1.040	0.257	0.135	0.796	$-0.095^{\dagger}$	0.941	0.299	0.691
Gemini Flash Lite	0.659	3.32	0.713	0.274	0.182	0.655	$+0.036$	0.942	0.277	0.693
LLaMA 3.1 8B	0.659	<u>14.13</u>	<u>1.782</u>	<u>0.116</u>	<u>0.142</u> *	<u>0.418</u>	$-0.048^{\dagger}$	<u>0.642</u>	<u>0.526</u>	0.849
GPT-5.4 Nano	0.696	2.57	0.841	0.227	0.155	0.568	$-0.092^{\dagger}$	0.676	0.160	<u>0.933</u>
LLaMA 4 Maverick	0.682	3.83	0.781	0.234	0.205	0.692	$-0.019^{\dagger}$	0.794	0.276	0.769

* Artifact of near-uniform posterior outputs, not genuine calibration.

[†] Oracle beliefs reduce performance vs. baseline (behavioral lock-in).

7.2 Experiment 1: Baseline Strategic Performance

Table 9: Experiment 1: Baseline strategic performance metrics. $\bar{R}^{\text{norm}}$ is normalized payoff $\in [0, 1]$; CR is cooperation rate; Total is mean cumulative episode payoff (raw, unnormalized). Best per metric **bolded**; worst underlined.

Metric	DeepSeek	Gem. 2.5P	Gem. FL	LLaMA 8B	GPT-Nano	LLaMA 4M
$\bar{R}^{\text{norm}}$ ↑	<u>0.634</u>	0.682	0.659	0.659	0.696	0.682
CR ↑	<u>0.480</u>	0.688	0.706	0.748	0.879	0.724
Total Payoff ↑	<u>25.07</u>	27.43	26.23	26.96	28.27	27.43

GPT-5.4 Nano achieves the highest overall normalized payoff ($\bar{R}^{\text{norm}} = 0.696$) and the highest total payoff (28.27 per episode) through near-universal cooperation (CR = 0.879). DeepSeek achieves the lowest overall payoff (0.634) but the highest Prisoner’s Dilemma payoff (0.436) through selective

Table 10: Experiment 1: Normalized payoff per game. 95% CIs shown in full in Appendix C, Table 20.

Game	DeepSeek	Gem. 2.5P	Gem. FL	LLaMA 8B	GPT-Nano	LLaMA 4M
Prisoner’s Dilemma	0.436	0.536	0.525	0.507	<u>0.520</u>	0.516
Stag Hunt	0.868	0.871	0.881	0.886	0.875	0.876
Chicken	<u>0.342</u>	0.424	0.329	0.486	0.523	0.444
Battle of Sexes	0.889	0.895	0.899	0.759	0.865	0.891
Overall	0.634	0.682	0.659	0.659	0.696	0.682

defection, at the cost of the worst Chicken performance (0.342). All models perform similarly in Stag Hunt (> 0.83) and Battle of the Sexes (> 0.75), where high cooperation aligns with good outcomes for the majority of opponent types. See Figure 1 for the two-axis characterization.

7.3 Experiment 2: Belief Quality and Calibration

Table 11: Experiment 2: Full belief quality and calibration metrics. KL and Brier computed on identifiable probes only ($\max_{\theta} P^* \geq 0.60$; N_{ident} shown). TypeAcc, ECE, MCE, and PredAcc computed on all 1,080 probes. Chance for TypeAcc = 0.111. **Bold** = best; underline = worst.

Metric	DeepSeek	Gem. 2.5P	Gem. FL	LLaMA 8B	GPT-Nano	LLaMA 4M
KL Div. $\downarrow$ (nats)	7.75	8.11	3.32	<u>14.13</u>	2.57	3.83
Brier Score $\downarrow$	1.030	1.040	0.713	<u>1.782</u>	0.841	0.781
Type Accuracy $\uparrow$	0.253	0.257	0.274	<u>0.116</u>	0.227	0.234
ECE $\downarrow$	0.211	0.135	0.182	0.142*	0.155	0.205
MCE $\downarrow$	0.289	0.164	0.253	<u>0.204</u> *	0.197	0.276
Pred. Accuracy $\uparrow$	0.900	0.881	0.876	0.897	0.906	<u>0.875</u>
N_{ident}	302	140	131	107	92	130

*Near-uniform outputs; ECE/MCE not indicative of genuine calibration.

Prediction accuracy vs. type accuracy. Prediction accuracy (one-step-ahead opponent action) and type accuracy ($\text{argmax posterior} = \theta^*$) measure different aspects of inference. GPT-5.4 Nano leads on prediction accuracy (0.906) but ranks fourth on type accuracy (0.227), while Gemini Flash Lite leads on type accuracy (0.274) with lower prediction accuracy (0.876). This dissociation reflects the difference between short-horizon action prediction (which a cooperation heuristic can achieve) and type identification (which requires distinguishing among all nine opponent types).

MCE reveals worst-case calibration failures. While ECE measures average calibration, MCE identifies the worst bin. Gemini 2.5 Pro achieves the lowest MCE (0.164), meaning even its most overconfident predictions are close to their empirical accuracy. DeepSeek’s MCE (0.289) reveals a substantially worse worst-case confidence tier despite moderate ECE (0.211).

Temporal KL dynamics across probe rounds $t \in \{3, 6, 10\}$ are shown in Figure 4; no model exhibits the expected monotone decrease.

Table 12: Experiment 3: Belief–action coupling metrics. Δ_{B-A} and Δ_{B2-A} are payoff gains of single-call and two-stage belief protocols over baseline. $BUC = \Delta_{C-A}$ is the oracle payoff gain. CAA = conditional action agreement on identifiable states. Coh(B) and Coh(B2) are EU-consistent coherence. **Bold** = best per column; underline = worst.

Metric	DeepSeek	Gem. 2.5P	Gem. FL	LLaMA 8B	GPT-Nano	LLaMA 4M
$\Delta_{B-A} \uparrow$	+0.022	+0.014	−0.007	<u>−0.062</u>	+0.022	+0.004
$\Delta_{B2-A} \uparrow$	−0.021	+0.007	−0.002	+0.011	+0.011	−0.005
BUC (Δ_{C-A})	+0.030	−0.095 [†]	+0.036	−0.048 [†]	−0.092 [†]	−0.019 [†]
CAA $\uparrow$	0.742	0.694	0.725	<u>0.044</u>	0.911	0.706
Coh(B) $\uparrow$	0.912	0.796	0.655	<u>0.418</u>	0.568	0.692
Coh(B2) $\uparrow$	0.769	0.763	0.619	0.534	0.555	0.671

[†]Oracle beliefs reduce performance (behavioral lock-in).

Table 13: Experiment 3: EU-consistent coherence Coh(B) per game, revealing Chicken as the universal diagnostic failure.

Model	PD	SH	Chicken	BoS
DeepSeek V3.2	0.889	0.948	0.844	0.967
Gemini 2.5 Pro	0.467	0.930	0.822	0.967
Gemini Flash Lite	0.215	0.982	0.452	0.970
LLaMA 3.1 8B	0.067	0.833	0.182	0.589
GPT-5.4 Nano	0.204	0.993	0.089	0.985
LLaMA 4 Maverick	0.226	0.974	0.589	0.978

7.4 Experiment 3: Belief–Action Coupling

CAA vs. Coh(B): consistency without rationality. GPT-5.4 Nano achieves the highest conditional action agreement (CAA = 0.911) but only moderate EU-coherence (Coh(B) = 0.568). This gap — high consistency, low rationality — is the defining failure mode: the model’s actions are self-consistent with its stated beliefs but those actions are not EU-optimal under those beliefs. The coupling hierarchy in Figure 5 makes this pattern visible across all six models.

B2 two-stage protocol. The B2 protocol (explicit belief articulation before action) helps only LLaMA 3.1 8B meaningfully, raising Coh from 0.418 to 0.534. For all other models, Coh(B2) is at most marginally different from Coh(B), and for DeepSeek it decreases (0.912 $\rightarrow$ 0.769). This rules out prompt-format as the cause of coupling failures: making beliefs explicit before action selection does not repair the pipeline.

7.5 Experiment 4: Posterior Intervention

Sensitivity without direction. LLaMA 3.1 8B and LLaMA 4 Maverick produce positive irrational gaps (+0.373 and +0.424 respectively): when these models react to a posterior change, they move in the wrong direction more often than they move in the right one. This is a stronger failure than non-responsiveness because it demonstrates that posterior signals actively misdirect action, not merely fail to guide it.

Discrete vs. continuous signal blindness. DirAcc(margin) is universally low (0.137–0.320) across all models, while DirAcc(flip) reaches 0.94. This gap reveals a structural failure: models respond to discrete EU-action flips but are blind to continuous decision-surface shifts. Figure 6 visualises this

Table 14: Experiment 4: Full causal intervention metrics. DirAcc(flip) = directional accuracy on EU-action-flip-valid probes. DirAcc(margin) = directional accuracy on margin-only-valid probes. Sens = any-action-change sensitivity. Irrational Gap = Sens − DirAcc(flip) (positive = active misdirection). **Bold** = best per column; underline = worst.

Model	DirAcc(flip) ↑	DirAcc(margin) ↑	Sens ↑	Irrat. Gap ↓	Valid N
DeepSeek V3.2	0.701	0.306	0.535	−0.166	1,866
Gemini 2.5 Pro	0.941	0.303	0.532	−0.409	1,832
Gemini Flash Lite	0.942	0.268	0.503	−0.439	1,793
LLaMA 3.1 8B	<u>0.642</u>	<u>0.137</u>	<u>0.269</u>	+0.373	1,833
GPT-5.4 Nano	0.676	0.320	0.512	+0.356	1,888
LLaMA 4 Maverick	0.794	0.194	0.370	+0.424	1,764

Table 15: Experiment 4: DirAcc(flip) and decision sensitivity (Sens) per game. PD has no flip-valid probes (dominant strategy); margin-only DirAcc is reported in its place.

Model	PD (margin only)		Stag Hunt		Chicken		BoS	
	DirAcc	Sens	DirAcc	Sens	DirAcc	Sens	DirAcc	Sens
DeepSeek V3.2	0.670	0.670	0.981	0.509	0.150	0.527	0.981	0.488
Gemini 2.5 Pro	0.758	0.758	0.978	0.472	0.859	0.508	0.996	0.490
Gemini Flash Lite	0.596	0.596	0.985	0.482	0.812	0.492	0.996	0.482
LLaMA 3.1 8B	0.151	0.151	1.000	0.315	0.067	0.082	0.913	0.495
GPT-5.4 Nano	0.812	0.812	0.996	0.484	0.129	0.347	0.964	0.529
LLaMA 4 Maverick	0.363	0.363	1.000	0.490	0.253	0.076	1.000	0.482

universal pattern. In the Prisoner’s Dilemma — the dominant-strategy game where flip-valid probes do not occur — Sensitivity ranges from 0.151 (LLaMA 8B) to 0.812 (GPT-5.4 Nano), reflecting each model’s responsiveness to margin changes alone.

7.6 Experiment 5: Cognitive Theory Metrics

Update magnitude and the rigidity paradox. GPT-5.4 Nano is the worst updater in the benchmark: mean update magnitude 0.362 (lowest) paired with the highest rigidity index (0.933). It forms the most accurate beliefs (JS = 0.160) but almost never revises them. This combination — accurate but frozen beliefs — is the structural correlate of the belief-action decoupling: the model infers the opponent correctly at $t = 3$, maintains that posterior nearly unchanged through $t = 10$, and acts independently of it throughout. Gemini Flash Lite shows the opposite: highest update magnitude (0.796) with moderate rigidity (0.693), suggesting active belief revision that partially translates into better deceptive-type identification (H6 Gap = 0.111, best).

Posterior consistency. Posterior consistency (mean JS between consecutive probe posteriors) measures internal stability of the belief system across time. GPT-5.4 Nano achieves the lowest value (0.069), confirming that its posteriors are highly self-consistent — but not because they are converging to the oracle, rather because they are near-identical across rounds (frozen). LLaMA 4 Maverick (0.132) and Gemini Flash Lite (0.191) show moderate stability with more active revision. DeepSeek (0.186) and LLaMA 8B (0.192) are nearly identical on this metric despite very different inference quality, suggesting

Table 16: Experiment 5: Full cognitive theory metrics. JS = mean JS divergence from oracle (H1). RI = Rigidity Index (H5, lower = more game-adaptive). H6 Gap = deceptive vs. non-deceptive type-accuracy gap. Upd. Mag = mean L1 belief update magnitude between consecutive probes (H7, higher = more responsive). Post. Cons. = mean JS between consecutive probe posteriors (lower = more internally stable). H2 Coh = EU-consistent coherence. H7 Corr = 0.000 for all models (omitted). **Bold** = best; underline = worst.

Metric	DeepSeek	Gem. 2.5P	Gem. FL	LLaMA 8B	GPT-Nano	LLaMA 4M
JS (H1) ↓	0.315	0.299	0.277	<u>0.526</u>	0.160	0.276
RI (H5) ↓	0.420	0.691	0.693	0.849	<u>0.933</u>	0.769
H6 Gap ↓	0.245	0.261	0.111	0.149	0.292	<u>0.301</u>
Upd. Mag ↑	0.706	0.776	0.796	0.557	<u>0.362</u>	0.644
Post. Cons. ↓	0.186	0.222	0.191	0.192	0.069	<u>0.132</u> [‡]
H2 Coh ↑	0.912	0.796	0.655	<u>0.418</u>	0.568	0.692
H7 Corr	0.000	0.000	0.000	0.000	0.000	0.000

[‡]Lower posterior consistency = more stable beliefs. LLaMA 4 Maverick’s low value reflects consistency but not accuracy.

posterior consistency is not a reliable proxy for belief accuracy.

Universal H7 failure. Every model returns $\rho_{H7} = 0.000$: prediction error magnitude and belief update magnitude are completely orthogonal. No model increases its update in response to higher surprisal. Combined with the update magnitude data above, this confirms that LLM beliefs are *snapshot representations* rather than continuously maintained and revised states.

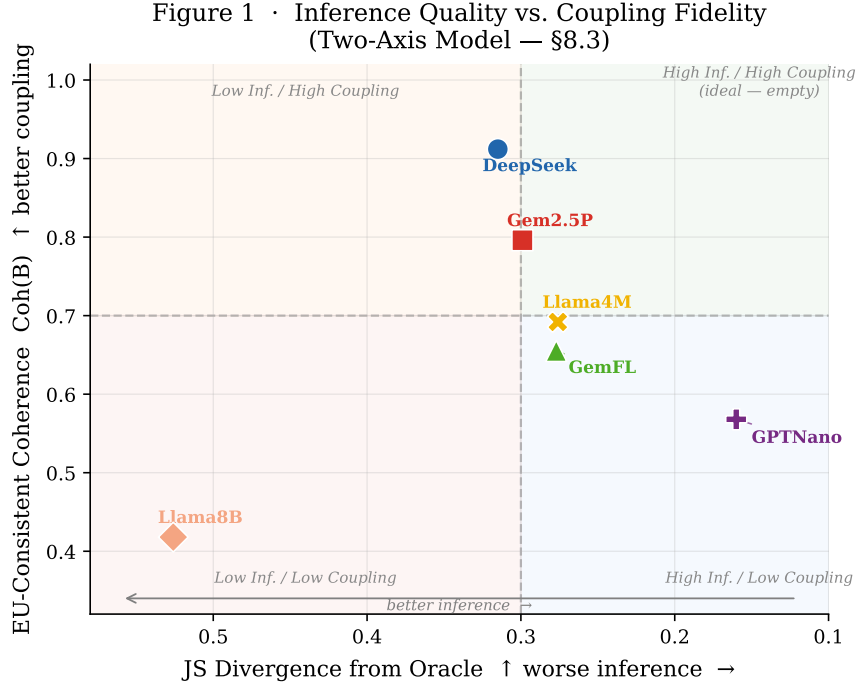


Figure 1: **Inference quality vs. coupling fidelity (two-axis model).** Each point is a model placed by its mean JS divergence from the oracle posterior (x-axis; lower = better inference, axis inverted) and its EU-consistent coherence Coh(B) (y-axis; higher = better coupling). Dashed lines mark the quadrant boundaries at $JS \approx 0.30$ and $Coh \approx 0.70$. The top-left ideal quadrant (high inference, high coupling) is empty for all evaluated models. GPT-5.4 Nano occupies the high-inference / low-coupling region: it forms near-Bayesian beliefs but does not act on them. DeepSeek V3.2 occupies the moderate-inference / high-coupling region: coarser beliefs with a tighter belief-to-action pathway. LLaMA 3.1 8B occupies the low-inference / low-coupling corner.

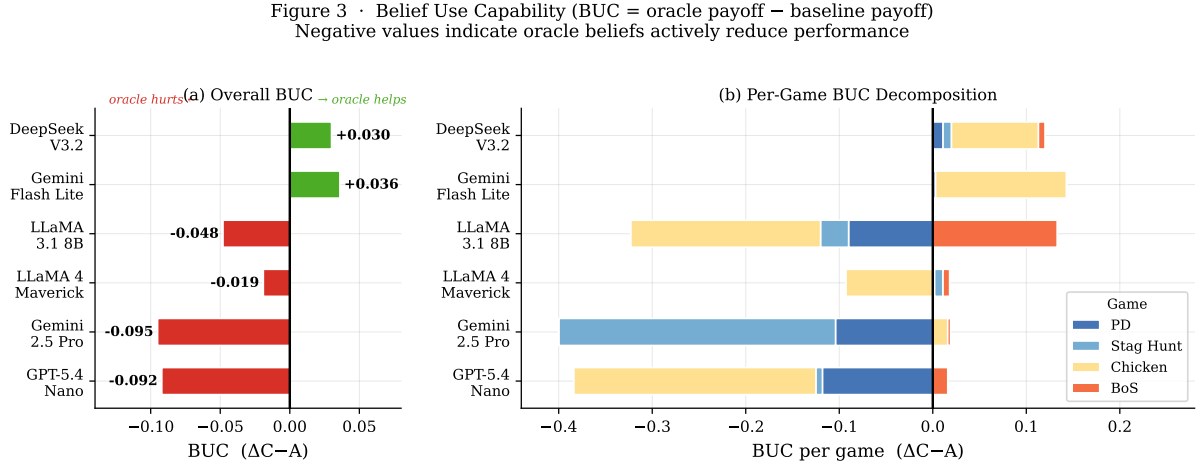


Figure 3: **Belief Use Capability (BUC = oracle payoff – baseline payoff).** *Left:* overall BUC per model; red bars extend left (oracle beliefs hurt performance), green bars right (oracle beliefs help). Four of six models are left of the zero line, falsifying the premise that accurate beliefs always improve outcomes. *Right:* per-game decomposition. Chicken is the only game where oracle beliefs consistently add value for the models with positive overall BUC (DeepSeek: +0.093; Gemini Flash Lite: +0.140), while Stag Hunt is where the largest negative values arise (Gemini 2.5 Pro: -0.296) because the baseline cooperative policy was already near-optimal there.

Figure 2 · EU-Consistent Coherence Coh(B) by Model and Game
(green = rational coupling · red = decoupled)

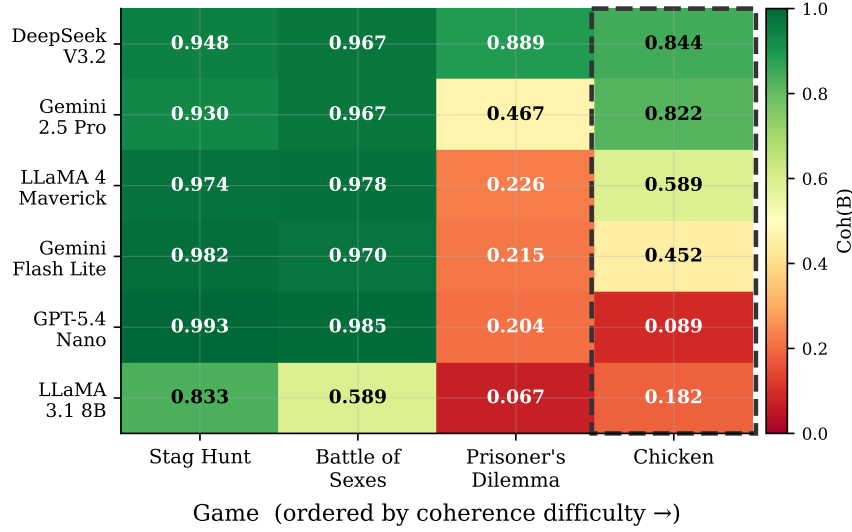


Figure 2: **EU-consistent coherence Coh(B) by model and game.** Cell colour encodes coherence (green = 1.0, red = 0.0; values printed in each cell). Games are ordered left to right by increasing coherence difficulty. The Chicken column (highlighted, dashed border) universally degrades across all six models, confirming it as the diagnostic game: GPT-5.4 Nano achieves Coh(B)= 0.993 in Stag Hunt but collapses to 0.089 in Chicken. No constant cooperation policy can approximate EU-optimality in Chicken, so this column isolates models that genuinely use beliefs from those relying on game-structure heuristics.

Figure 4 · Belief Convergence Dynamics Across Probe Rounds
(Bayesian convergence predicts monotone decrease — no model shows it)

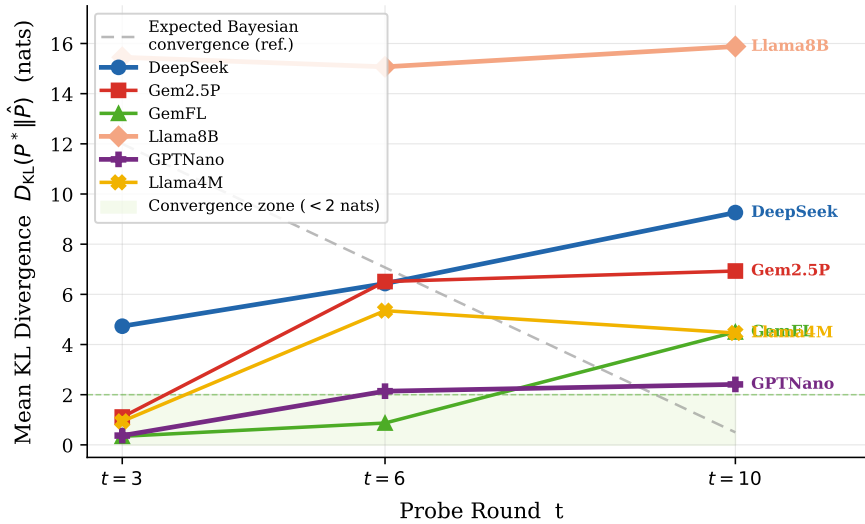


Figure 4: **Belief convergence dynamics across probe rounds $t \in \{3, 6, 10\}$.** A Bayesian agent should show monotonically decreasing KL divergence from the oracle posterior as evidence accumulates (dashed reference line). No model follows this trajectory. GPT-5.4 Nano starts lowest (0.37 nats at $t = 3$) but increases to 2.41 by $t = 10$. DeepSeek increases monotonically (4.73 $\rightarrow$ 6.43 $\rightarrow$ 9.26 nats). Gemini Flash Lite shows a sharp late-round spike. LLaMA 3.1 8B remains near the theoretical ceiling (≈ 16 nats) at all three rounds. The shaded green band marks the convergence zone (< 2 nats).

Figure 5 · Belief-Action Coupling Hierarchy (L1 → L4)
A model can pass lower levels while failing higher ones

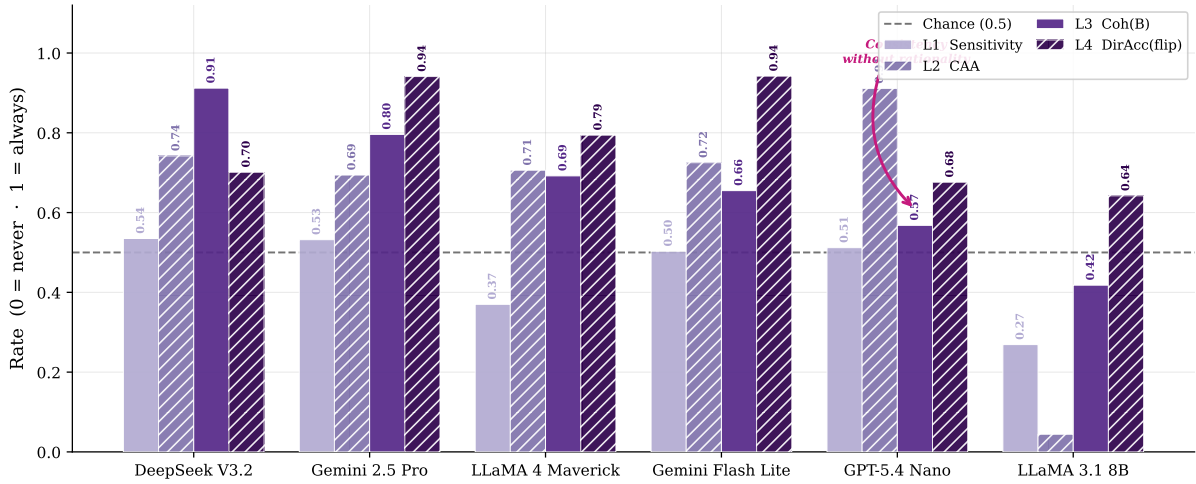


Figure 5: **Belief-action coupling hierarchy across all six models.** Four bars per model show performance at each level of the diagnostic hierarchy: L1 Sensitivity (does the model respond to any posterior change?), L2 CAA (do expressed beliefs align with action choices?), L3 Coh(B) (are actions EU-optimal under stated beliefs?), L4 DirAcc(flip) (does causally injected belief produce EU-optimal action?). The dashed line marks chance (0.5). The annotated arrow on GPT-5.4 Nano highlights the “consistency without rationality” failure: high L2 CAA (0.911) but only moderate L3 Coh(B) (0.568), meaning the model is self-consistent but not EU-rational. Gemini models show the opposite inversion: L4 DirAcc exceeds L3 Coh, meaning causal injection works better than the model’s own coupling.

Figure 6 · Discrete vs. Continuous Posterior Signal Response
All models respond to action-flip probes but are blind to continuous EU-margin shifts

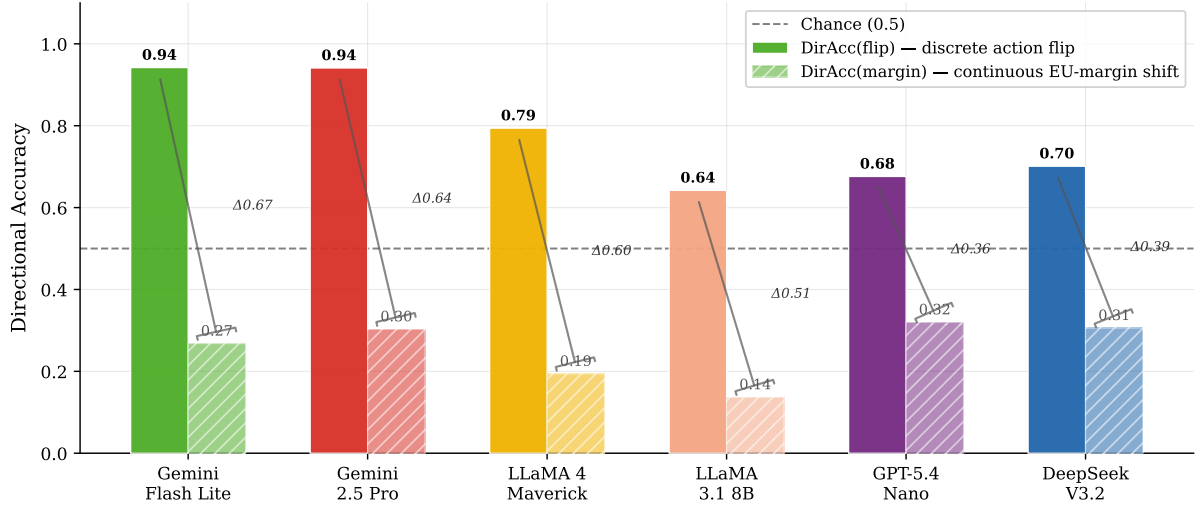


Figure 6: **Discrete vs. continuous posterior signal response.** Solid bars show DirAcc(flip): how often the model selects the EU-optimal action when a counterfactual posterior prescribes a discrete action change ($C \rightarrow D$ or $D \rightarrow C$). Hatched bars show DirAcc(margin): how often the model moves in the correct direction when the posterior shift changes the EU margin without crossing the action threshold. The gap Δ (annotated per model) is universal and large (0.36–0.67). All models are switch-responsive but signal-blind: they follow discrete belief flips with high fidelity (0.64–0.94) but fail to respond to continuous decision-surface shifts (0.14–0.32). This reveals that belief-to-action computation operates as a binary threshold rather than a continuous expected utility signal.

Figure 7 · Does Adding Belief Information Improve Outcomes?
A rational agent's line should rise monotonically $A \rightarrow B \rightarrow B2 \rightarrow C$

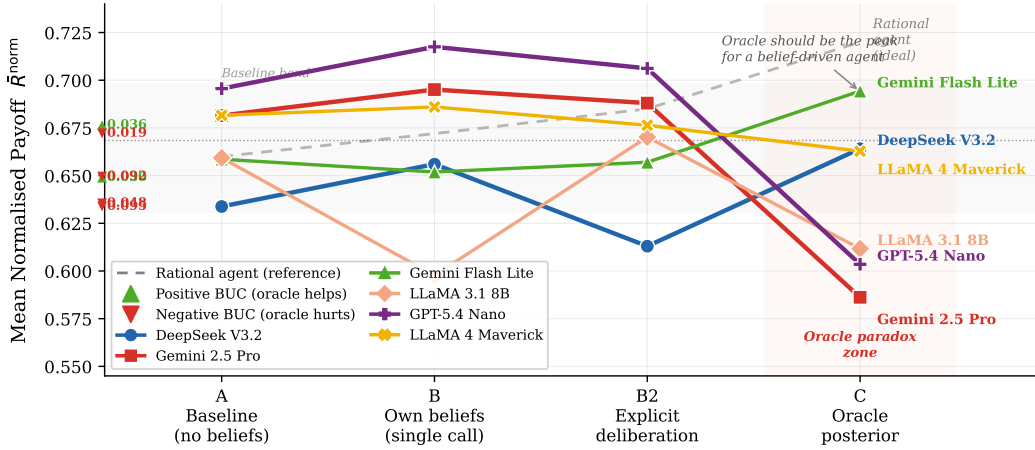


Figure 7: **The value of belief information across four experimental conditions.** Each line traces one model’s normalised payoff from condition A (baseline, no beliefs) through B (own elicited beliefs, single call), B2 (explicit two-stage deliberation), and C (oracle ground-truth posterior). The dashed grey reference shows the monotonically increasing trajectory a rational belief-driven agent must follow: more accurate beliefs should always improve outcomes. No model follows this trajectory. Four of six lines *fall* at condition C relative to their baseline A (highlighted oracle paradox zone, red shading), meaning ground-truth beliefs actively hurt performance — the direct empirical signature of belief–action decoupling. Only DeepSeek V3.2 and Gemini Flash Lite finish above their baselines at C, and both do so exclusively because Chicken forces belief-conditioned reasoning. The left-margin arrows show the net BUC (green $\triangle$ = oracle helps, red ∇ = oracle hurts) per model.

7.7 Cross-Experiment Synthesis

The five experiments produce 18 preregistered metrics across 6 models, 4 games, and 9 opponent types. No single metric tells the full story; the findings below emerge only when results are read jointly across all experiments. Figures 1–6 provide visual support for each finding.

Finding 1: Belief quality and coupling fidelity are empirically dissociable. GPT-5.4 Nano achieves the best inference quality by every metric — lowest KL (2.57 nats), lowest JS (0.160), highest prediction accuracy (0.906), most stable posteriors (consistency 0.069) — yet its EU-consistent coherence is only moderate (0.568), its BUC is negative (−0.092), and its update magnitude is the lowest in the benchmark (0.362). Its beliefs are accurate and frozen; they are not consulted when actions are selected. DeepSeek V3.2 is the exact mirror: substantially worse inference (KL 7.75, JS 0.315) yet the highest coherence (0.912), the only positive BUC (+0.030), and the most game-adaptive behaviour (RI 0.420). Forming accurate beliefs and acting on them are architecturally independent tasks. No model occupies the high-inference, high-coupling ideal quadrant (Figure 1), confirming the failure is structural, not a matter of scale or capability.

Finding 2: The oracle posterior paradox exposes behavioral lock-in. Four of six models perform *worse* with oracle-accurate beliefs than with their own inferred beliefs (Table 12). Gemini 2.5 Pro loses the most overall (BUC = −0.095) and the most in a single game (Stag Hunt: −0.296), where its baseline cooperative policy was already near-optimal against the majority of opponent types. GPT-5.4 Nano loses −0.259 in Chicken alone. These losses are not oracle failures — they are the signature of a cooperation heuristic so deeply trained that it overrides any belief-conditioned signal that deviates from it. The only game where oracle beliefs consistently help is Chicken (DeepSeek +0.093, Gemini Flash +0.140) — precisely the game where no constant cooperation policy is near-optimal (Figure 3). Behavioral lock-in

and belief-action decoupling are therefore not independent failures; a model without a genuine coupling pathway will be actively harmed by accurate beliefs.

Finding 3: Chicken is the universal diagnostic game. All six models achieve $\text{Coh}(\text{B}) > 0.83$ in Stag Hunt but degrade substantially in Chicken (minimum drop: 0.104, DeepSeek; maximum drop: 0.904, GPT-5.4 Nano from 0.993 $\rightarrow$ 0.089) — worse than random (Table 13, Figure 2). Chicken is the only game in the benchmark where no constant action policy approximates EU-optimality: the correct action against `always_defect` (swerve) is the opposite of the correct action against `always_cooperate` (go straight), and the choice flips again against `tit_for_tat` and `gradual_defector`. Any model achieving high $\text{Coh}(\text{B})$ in Chicken is, by construction, using its posterior. Any model that cooperates by default will be anti-coherent there. Coherence in Stag Hunt and Battle of the Sexes is observationally indistinguishable from a cooperation heuristic; Chicken makes the distinction empirically testable. We recommend its inclusion as a mandatory game in future strategic LLM benchmarks.

Finding 4: Universal zero surprisal–update coupling reveals static belief representations. Every model returns $\rho_{H7} = 0.000$ — not near-zero, but structural zero, arising because all models express near-constant confidence regardless of prediction correctness. No model increases its update magnitude following a more surprising outcome. The GPT-5.4 Nano extreme (update magnitude 0.362, posterior consistency 0.069) confirms near-identical posteriors across all three probe rounds despite the episode evolving. DeepSeek and Gemini Flash Lite update more actively (0.706 and 0.796) but still with constant expressed confidence. Across all six models, expressed confidence is a linguistic token generated to fill a schema field, not a summary of the model’s internal uncertainty. Beliefs are re-inferred from full context at each probe round, not revised from prior estimates — making the Bayesian update concept architecturally inapplicable to these models as deployed.

Finding 5: The coupling hierarchy is non-monotonic. Two distinct failure patterns emerge from the four-level hierarchy (Figure 5). *Consistency without rationality* (GPT-5.4 Nano): high CAA (0.911, L2) but only moderate $\text{Coh}(\text{B})$ (0.568, L3). Its actions are self-consistent with its stated beliefs, but those actions are not EU-optimal under those beliefs — both are produced by the same cooperation prior rather than one following from the other. *Causal responsiveness without self-coupling* (both Gemini models): $\text{DirAcc}(\text{flip})$ (0.941–0.942, L4) substantially exceeds $\text{Coh}(\text{B})$ (0.655–0.796, L3). When an external posterior is injected, these models follow it with 94% fidelity; when reasoning from their own elicited posteriors, coherence is lower. The coupling infrastructure exists but the self-generated beliefs do not interface with it correctly — a narrower and more tractable failure than GPT-5.4 Nano’s complete decoupling.

Overall model characterisations. The five findings above support the following cross-cutting characterisations of each model:

GPT-5.4 Nano is the best inference engine in the benchmark but the most rigid strategist. It forms near-Bayesian posteriors, maintains them with high internal consistency, and then acts almost entirely through a cooperation heuristic. The belief and action pipelines are empirically decoupled.

DeepSeek V3.2 is the most rational strategist in the benchmark given its beliefs. Its posteriors are coarser than GPT-5.4 Nano’s, but it acts on them: the belief-to-action pathway is the tightest among all six models. It is the only model with consistently positive BUC, the lowest rigidity index, and the best EU-consistent coherence.

Gemini 2.5 Pro is the best-calibrated model: its expressed confidence most accurately tracks its prediction accuracy (ECE 0.135, MCE 0.164). It responds correctly to externally injected posteriors ($\text{DirAcc}(\text{flip})$ 0.941), but loses the most from oracle injection in Stag Hunt (−0.296), exposing the deepest behavioral lock-in against cooperative opponents.

Gemini Flash Lite achieves the best type identification accuracy (0.274), the lowest deceptive-type gap (0.111), and the highest update magnitude (0.796). Its beliefs are the most discriminative and the most actively revised. It is the model most resilient to deception — a property that may reflect training on adversarial interaction data.

LLaMA 4 Maverick performs competitively on payoff and causal responsiveness (DirAcc(flip) 0.794) but exhibits the largest irrational gap (+0.424): when it reacts to a posterior change, it moves in the wrong direction more than in the right one. Its coupling is active but misdirected.

LLaMA 3.1 8B shows near-complete inference failure across every metric: type accuracy at chance baseline (0.116), Brier score near ceiling (1.782), and CAA of 0.044 (essentially random action-belief alignment). It is the only model where the B2 protocol provides a meaningful improvement, suggesting that explicit prior articulation partially compensates for an otherwise dysfunctional inference mechanism.

8 Discussion

8.1 Revisiting the Central Thesis

This paper opened with a causal question: do LLM actions depend on LLM beliefs, or are the two parallel outputs of a shared heuristic that happen to correlate under standard conditions? The five experiments provide a precise and consistently negative answer. Stated beliefs routinely diverge from the oracle posterior. Even when beliefs are accurate, EU-consistent coherence in the games that require belief-conditioned reasoning sits well below the 1.0 that rational choice demands. Models respond to causally injected posteriors sensitively but not rationally. And four of six models perform worse with oracle beliefs than without them.

Taken together, these results falsify the view that strong behavioral performance implies coherent belief-driven reasoning. The models evaluated here are not rational agents with imperfect beliefs. They are agents whose actions are governed primarily by game-structure heuristics — cooperation defaults, defection in the Prisoner’s Dilemma, coordination in symmetric games — with beliefs functioning as a secondary representation that is largely disconnected from the action-selection mechanism. The specific evidence is documented in §7.7; this section interprets what that evidence means about the architecture of LLM strategic cognition.

8.2 What the Findings Reveal About LLM Architecture

Belief quality and coupling are architecturally independent. The dissociation between GPT-5.4 Nano (best inference, moderate coupling) and DeepSeek (moderate inference, best coupling) is not a capability gap — it is a structural one. Forming an accurate posterior requires learning to aggregate evidence from interaction history into a probability simplex. Acting rationally on a posterior requires a separate computation: evaluating expected utility under that distribution and returning its argmax. These two operations can — and apparently do — run in entirely separate computational pathways in current models.

What determines whether the coupling pathway exists? The most likely answer is training signal. A model trained with reinforcement feedback defined over payoff outcomes alone has no gradient pressure to use its expressed beliefs in its action computation. The belief distribution is elicited and evaluated by the evaluator; the action is what the model is rewarded for. Without a training signal that explicitly rewards actions that are EU-consistent with the expressed posterior, the two outputs will remain statistically independent. This has a direct methodological implication: outcome-based evaluation — the dominant paradigm in LLM benchmarking — is blind to this failure mode by construction.

Behavioral lock-in as a training artifact. The oracle posterior paradox (four of six models perform worse with accurate beliefs) is not a failure of the oracle. It is the signature of a training process that has optimised cooperation policies against a population of opponents in which cooperation is near-optimal for the majority. When an oracle belief correctly identifies an adversarial opponent and prescribes defection, the cooperative behavioral policy that training has installed overrides it — because that policy produces higher expected payoff in expectation over the training distribution. The model cannot exploit the oracle because the mechanism that would do so — a genuine EU computation — was never trained into the coupling pathway.

The implication is that oracle-injection experiments (Experiment 3C) serve as a diagnostic for the *existence* of a coupling pathway. A model with $BUC \approx 0$ either has near-perfect inference or has a functional coupling pathway. A model with large negative BUC has a functional cooperation heuristic and no coupling pathway at all. This makes BUC not just a payoff metric but a structural probe.

Chicken reveals what cooperative games conceal. The universal collapse of EU-coherence in the Chicken game is not just a finding about Chicken — it is a methodological warning about benchmarks that rely exclusively on cooperative games. In Stag Hunt and Battle of the Sexes, any model that cooperates consistently will appear coherent, because cooperation is the EU-optimal action under the vast majority of posteriors. Coherence in those games is observationally indistinguishable from the cooperation heuristic. Chicken destroys this equivalence: there is no constant action that is approximately EU-optimal across all nine opponent types, so coherence requires genuine posterior-conditioned choice. GPT-5.4 Nano’s coherence of 0.993 in Stag Hunt and 0.089 in Chicken is not a paradox — it is the empirical signature of a cooperation prior masquerading as rational inference in one game and failing in the other.

Static belief snapshots and the absence of predictive processing. The universal $\rho_{H7} = 0.000$ reveals something deeper than miscalibration. It reveals that expressed confidence is not a summary of the model’s internal uncertainty state — it is a linguistic token generated to fill a schema field. If confidence were connected to any internal inference signal, there would be at least weak positive correlation between expressed prediction error and posterior update magnitude. The zero correlation means this connection does not exist.

The architectural consequence is that LLM belief systems are *stateless across turns* in a way that Bayesian or approximately-Bayesian systems are not. The posterior at $t = 10$ is computed from the full episode history as a fresh inference, not by updating the posterior from $t = 6$. This is consistent with the transformer architecture: each forward pass has access to the full context window, so there is no representational need to maintain a running belief state. But it means that the notion of “belief updating” — central to Bayesian rationality and to the H7 hypothesis — is architecturally inapplicable to current LLMs as deployed here. They do not update beliefs; they re-infer them.

8.3 The Two-Axis Model of LLM Strategic Cognition

The results support characterising LLM strategic behaviour along two empirically independent axes:

- **Inference quality:** how accurately the model tracks the Bayesian posterior over opponent types (measured by KL, JS, Brier, TypeAcc).
- **Coupling fidelity:** the degree to which inferred beliefs causally determine action selection (measured by Coh(B), DirAcc, BUC, update magnitude).

Models occupy distinct quadrants of this space (Figure 1). GPT-5.4 Nano: high inference, low coupling. DeepSeek V3.2: moderate inference, high coupling. LLaMA 3.1 8B: low inference, low coupling. No model clearly occupies the high-inference, high-coupling quadrant that would characterise a genuinely rational strategic agent.

This taxonomy has direct diagnostic value. A model in the GPT-5.4 Nano quadrant has a functioning inference engine and a broken coupling pathway. The intervention is targeted: training on EU-consistent feedback, explicit expected-utility computation in the reasoning chain, or belief-conditioned action labels that force the expressed posterior to serve as input to action selection. A model in the LLaMA 3.1 8B quadrant requires improvements to both components: better in-context inference and coupling consistency. A model in the DeepSeek quadrant already has a functional coupling pathway; the intervention is to improve inference quality so that the pathway is fed accurate beliefs.

Crucially, the two axes are not correlated with model size or provider family. The largest gap between inference quality and coupling fidelity is found in GPT-5.4 Nano — one of the strongest models in the benchmark. This confirms that the failure is not resolved by scale alone and requires targeted architectural or training intervention.

8.4 Implications for LLM Agent Design

Evaluation methodology. Our results establish that behavioral performance—payoff, win rate, task success—is an insufficient indicator of coherent reasoning in partially observable settings. A benchmark that measures only outcomes will rank GPT-5.4 Nano first and miss its fundamental belief–action decoupling. Future LLM agent evaluations should explicitly instrument the inference–representation–action pipeline by including: (i) structured belief elicitation at intermediate timesteps; (ii) oracle-injected conditions to bound the value of accurate beliefs; and (iii) posterior intervention probes to test causal belief–action dependence. These additions are cheap relative to full episode evaluation—the three probe calls per episode in our design add 30% overhead to Stage 1—but reveal dimensions of reasoning invisible to outcome measurement alone.

Training for belief-conditioned action. The belief–action dissociation we observe is unlikely to be resolved by scale alone. GPT-5.4 Nano is already among the most capable models in the benchmark and shows the strongest belief quality—yet its coupling is moderate. What appears missing is not reasoning capacity but a mechanism that *uses* expressed beliefs as inputs to action computation. Reinforcement learning from belief-conditioned feedback—rewarding actions that are EU-optimal under the stated posterior, not just actions that yield good outcomes—could directly target the coupling dimension. The B2 two-stage protocol in our benchmark provides a natural training signal: generate beliefs, then select the action that maximizes expected payoff under those beliefs, and evaluate EU-consistency rather than payoff alone.

Deceptive opponents as a ToM benchmark. The deceptive-type accuracy gap (H6) provides a lightweight proxy for theory of mind depth in a strategic setting. Models that cannot identify `deceptive_opportunists` and `gradual_defector` are attributing strategic cooperation to genuine cooperative intent rather than instrumental deception—a failure of second-order belief attribution that maps directly onto classical ToM measures. The fact that Gemini Flash Lite (Gap = 0.111) substantially outperforms GPT-5.4 Nano (0.292) on this metric, despite GPT-5.4 Nano having better overall inference quality, suggests that deceptive type identification is a semi-independent capability that may require targeted training on adversarial interaction data.

8.5 Limitations

Verbalized beliefs may not equal internal beliefs. The most fundamental limitation of this framework is that the posteriors we evaluate are *verbalized*—they are outputs of a generation process that may or may not reflect the model’s internal computational state. A model could achieve perfect verbalized calibration while computing actions through a wholly separate mechanism, or could exhibit high EU-coherence purely through the consistency of its language output rather than genuine belief-driven computation. Mechanistic interpretability methods—activation patching, causal tracing [Meng et al., 2022]—are

necessary complements to our behavioral evaluation, and we regard our metrics as necessary but not sufficient evidence for or against genuine belief-driven reasoning.

Ten-round episodes limit type identifiability. With $T = 10$ rounds and nine opponent types, many episodes terminate before the oracle posterior reaches confident identifiability ($\tau = 0.60$). Only 302 of 1,080 probe observations cross this threshold, meaning that roughly 72% of probe states are evidence-poor from the oracle’s perspective. This limits statistical power for belief quality metrics and may understate the inference capability of stronger models that converge quickly to concentrated posteriors. Longer episodes (e.g. $T = 20$ – 30) would provide richer identifiability but at substantially higher API cost.

Temperature zero and deterministic decoding. All experiments use temperature = 0.0. Deterministic decoding maximizes reproducibility but may suppress stochastic reasoning strategies that could improve belief calibration or coupling under sampling. Whether the dissociation between belief quality and coupling persists under stochastic decoding is an open empirical question; we regard it as a robustness check that future work should conduct.

No fine-tuned or instruction-tuned-for-games models. All models are evaluated in their released chat/instruction-tuned configurations without any game-specific fine-tuning. It is possible that targeted fine-tuning on repeated-game interaction data—or reinforcement learning from EU-consistent feedback—would substantially close the belief–action gap. Our results characterize the *default* capability of current models, not an upper bound on achievable performance.

The oracle posterior is an approximation. The oracle posterior $P^*(\theta \mid h_t)$ is itself computed under modeling assumptions: a uniform prior, a likelihood noise floor of $\varepsilon_{\text{lik}} = 0.01$, and additive smoothing at $\varepsilon_s = 10^{-6}$. For stochastic opponent types, the likelihood model is exact by construction; for deterministic types, the noise floor introduces a deliberate approximation. An oracle with zero noise floor would assign probability zero to non-matching actions after a single observation, making the posterior degenerate after the first round for deterministic types. The noise floor is thus a principled regularization, but it means our oracle is not the true Bayesian posterior—it is a regularized approximation that is nonetheless far more accurate than any model’s elicited beliefs.

No cross-episode memory. Each episode is initialized with an empty history. Models do not accumulate experience across episodes and cannot form beliefs about the distribution of opponent types in the benchmark. This is intentional for evaluating within-episode inference but prevents evaluation of meta-level learning—whether models become better at identifying opponent types over repeated exposures to the same type—which is a distinct and important capability not addressed by this benchmark.

8.6 Future Directions

Several extensions of this framework are well-motivated by the current results. First, extending the opponent type set to include mixed-strategy or learning opponents—types whose policies evolve during the episode in response to the LLM’s behavior—would stress-test belief dynamics in settings where the latent type is itself non-stationary, requiring online belief revision rather than simple posterior accumulation.

Second, the B2 two-stage protocol demonstrated a modest improvement in EU-coherence for weaker models. A natural follow-up is to explore whether an iterative refinement loop—generate beliefs, compute EU, check coherence, revise action if inconsistent—can be engineered at prompt-time to close the coupling gap without model fine-tuning.

Third, the deceptive-type gap suggests that identifying strategically deceptive agents is a meaningful and underexplored LLM capability. Extending the benchmark to include longer deception windows,

higher-order deception (opponents that adapt their deception strategy based on observed model responses), and larger type sets would provide a more complete characterization of ToM-depth capabilities.

Finally, the universal zero surprisal–update coupling warrants direct investigation through mechanistic interpretability. If posterior representations at consecutive probe rounds are genuinely independent computations rather than successive revisions, this should be detectable as the absence of any cross-timestep information flow in the model’s key–value cache or attention patterns—a prediction that activation patching experiments could test directly.

9 Conclusion

The question motivating this work is deceptively simple: when a large language model plays a strategy, does it do so because it *believes* something about its opponent? Answering this required moving beyond behavioral measurement entirely. We built a benchmark that instruments the inference–representation–action pipeline at each link, instantiated across four canonical 2×2 games, nine opponent policy types, and six frontier models, with an offline oracle Bayesian posterior as the normative ground truth against which every stated belief is scored. The result is the first evaluation framework that separates a model that reasons from one that merely approximates reasoning from the outside.

What we found was not a spectrum of belief-driven reasoning at different quality levels—it was, in most cases, the near-absence of a belief-to-action pathway. Models form beliefs and then largely discard them when selecting actions. EU-consistent coherence hovers near chance in the games where beliefs genuinely determine the optimal action. Oracle posteriors hurt four of six models, exposing deep behavioral lock-in. Injected counterfactual beliefs produce action changes that are sensitive but misdirected. And across every model, prediction error is completely orthogonal to belief update magnitude—no model updates more when surprised. The central claim of this paper is therefore a falsification: *strong behavioral performance in strategic interaction does not imply, predict, or correlate with coherent belief-driven reasoning in current large language models.*

This matters well beyond the specific games studied here. LLMs are increasingly deployed in settings structurally identical to this benchmark—environments with hidden agent types, partial observability, and decisions that should be conditioned on inferred latent state. A model that achieves high task success through heuristics correlated with good outcomes in training is brittle precisely when it matters most: novel opponents, adversarial environments, and distribution shifts where the heuristic no longer aligns with the right action.

The two-axis characterization that emerges—inference quality on one axis, coupling fidelity on the other—provides a practical roadmap. No model in this evaluation occupies the high-inference, high-coupling quadrant of a genuinely rational strategic agent. Reaching it will require training regimes that explicitly reward belief-conditioned action: reinforcement signals defined over EU consistency, not payoff alone, and evaluation protocols that penalize coherence failure, not just outcome failure. The benchmark introduced here provides the measurement infrastructure to track progress toward that goal. The work of building agents that reason from their beliefs rather than describe reasoning they do not perform begins with the ability to tell the difference.

References

Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9:1380–1390, 2025. doi: 10.1038/s41562-025-02172-y.

- Chris L. Baker, Rebecca R. Saxe, and Joshua B. Tenenbaum. Bayesian theory of mind: Modeling joint belief-desire attribution. In *Proceedings of the 33rd Annual Conference of the Cognitive Science Society*, pages 2469–2474, 2011.
- Marcel Binz and Eric Schulz. Turning large language models into cognitive models. *arXiv preprint arXiv:2306.03917*, 2023a. URL <https://arxiv.org/abs/2306.03917>.
- Marcel Binz and Eric Schulz. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023b. doi: 10.1073/pnas.2218523120.
- Colin F. Camerer. *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press, Princeton, NJ, 2003.
- Zhuang Chen, Jingzhe Shi, Wenhan Liu, Yiwen Wan, Zhichen Zhou, Fei Zhang, Yu Jian, Siyu Si, Tong Liu, Daiyuan Peng, et al. ToMBench: Benchmarking theory of mind in large language models. *arXiv preprint arXiv:2402.15052*, 2024. URL <https://arxiv.org/abs/2402.15052>.
- Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3):181–204, 2013. doi: 10.1017/S0140525X12000477.
- Anthony Costarelli, Mat Allen, Roman Hauksson, Grace Sodunke, Suhas Hariharan, Carlson Cheng, Wenjie Li, and Arjun Yadav. GameBench: Evaluating strategic reasoning abilities of LLM agents. *arXiv preprint arXiv:2406.06613*, 2024. URL <https://arxiv.org/abs/2406.06613>.
- Persi Diaconis and Donald Ylvisaker. Conjugate priors for exponential families. *The Annals of Statistics*, 7(2):269–281, 1979. doi: 10.1214/aos/1176344610.
- Jinhao Duan, Renming Zhang, James Diffenderfer, Bhavya Kailkhura, Lichao Sun, Elias Stengel-Eskin, Mohit Bansal, Tianlong Chen, and Kaidi Xu. GTBench: Uncovering the strategic reasoning limitations of LLMs via game-theoretic evaluations. *Advances in Neural Information Processing Systems*, 37, 2024. URL <https://arxiv.org/abs/2402.12348>. NeurIPS 2024.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall / CRC, New York, 1993. doi: 10.1007/978-1-4899-4541-9.
- Jonathan St. B. T. Evans. In two minds: Dual-process accounts of reasoning. *Trends in Cognitive Sciences*, 7(10):454–459, 2003. doi: 10.1016/j.tics.2003.08.012.
- John H. Flavell. Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, 34(10):906–911, 1979. doi: 10.1037/0003-066X.34.10.906.
- Gabriel Freedman, Francesca Toni, et al. Do LLMs act like rational agents? measuring belief coherence in probabilistic decision making. *arXiv preprint arXiv:2602.06286*, 2026. URL <https://arxiv.org/abs/2602.06286>.
- Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2): 127–138, 2010. doi: 10.1038/nrn2787.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. Causal abstraction for faithful model interpretation. *Journal of Artificial Intelligence Research*, 81:1–46, 2024. doi: 10.1613/jair.1.16226.
- Gerd Gigerenzer and Daniel G. Goldstein. Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review*, 103(4):650–669, 1996. doi: 10.1037/0033-295X.103.4.650.
- Tilman Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings*

- of *Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL <https://arxiv.org/abs/1706.04599>.
- John J. Horton, Apostolos Filippas, and Benjamin S. Manning. Large language models as simulated economic agents: What can we learn from homo silicus? *arXiv preprint arXiv:2301.07543*, 2023. URL <https://arxiv.org/abs/2301.07543>. NBER Working Paper No. 31122.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. URL <https://arxiv.org/abs/2207.05221>.
- Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.
- Michal Kosinski. Evaluating large language models in theory of mind tasks. *Proceedings of the National Academy of Sciences*, 121(45), 2024. doi: 10.1073/pnas.2405602121.
- Brenden M. Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2879–2888. PMLR, 2018.
- Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, Sameer Singh, Subbarao Kambhampati, and Chris Callison-Burch. Entity-based knowledge conflicts in question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7492–7503, 2021. URL <https://arxiv.org/abs/2109.05052>.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372, 2022. URL <https://arxiv.org/abs/2202.05262>.
- Meta AI. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, pages 2901–2907. AAAI Press, 2015.
- OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2024. URL <https://arxiv.org/abs/2303.08774>.
- Martin J. Osborne and Ariel Rubinstein. *A Course in Game Theory*. MIT Press, Cambridge, MA, 1994.
- Steve Phelps and Yvan Russell. Strategic intelligence in large language models: Evidence from evolutionary game theory. *arXiv preprint arXiv:2507.02618*, 2025. URL <https://arxiv.org/abs/2507.02618>.
- David Premack and Guy Woodruff. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4):515–526, 1978. doi: 10.1017/S0140525X00076512.
- Leonard J. Savage. *The Foundations of Statistics*. Wiley, New York, 1954.
- Herbert A. Simon. A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1):99–118, 1955. doi: 10.2307/1884852.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Graham Keeling, Albert Zeyer, et al. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2024. URL <https://arxiv.org/abs/2312.11805>.

- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965, 2023. URL <https://arxiv.org/abs/2305.04388>.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2203.11171>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022. URL <https://arxiv.org/abs/2201.11903>.
- Heinz Wimmer and Josef Perner. Beliefs about beliefs: Representation and constraining function of wrong beliefs in young children’s understanding of deception. *Cognition*, 13(1):103–128, 1983. doi: 10.1016/0010-0277(83)90004-5.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. *arXiv preprint arXiv:2306.13063*, 2024. URL <https://arxiv.org/abs/2306.13063>.

A Prompt Templates

All prompts are reproduced verbatim from `prompts.py`. Placeholders in angle brackets (e.g. `<game_name>`) indicate fields populated at runtime. The system prompt is cached by all providers supporting prompt caching and sent once per episode; user prompts are generated fresh at each round.

We use the `listings` package with a custom style to render all prompt blocks in a monospaced box exactly as they are submitted to the API.

A.1 Stage 1 & 2 — Shared System Prompt

Used for Experiments 1, 2, 3-A, 3-B, and 3-B2. Instantiated once per game per episode (≈ 350 tokens).

```
You are a strategic JSON-only reasoning agent playing a
repeated 2x2 game.
Your entire response must be a single JSON object.
No explanation text.

GAME: <game_name>
<description>

PAYOFF MATRIX (Your action, Opponent action ->
               your payoff, opponent payoff):
  You=C, Opponent=C -> you earn <cc0>, opponent earns <cc1>
  You=C, Opponent=D -> you earn <cd0>, opponent earns <cd1>
  You=D, Opponent=C -> you earn <dc0>, opponent earns <dc1>
  You=D, Opponent=D -> you earn <dd0>, opponent earns <dd1>

KNOWN OPPONENT TYPES: always_cooperate, always_defect,
  tit_for_tat, grim_trigger, win_stay_lose_shift,
  noisy_tit_for_tat, deceptive_opportunist,
  gradual_defector, adaptive_mirror
```

```
Your goal: infer the opponent's type from interaction
history and maximize your cumulative payoff.
```

RESPONSE FORMATS:

- Action only:


```
{"action": "C"} or {"action": "D"}
```
- Belief probe:


```
{"action": "C"|"D",
  "predicted_next_action": "C"|"D",
  "confidence": 0.0-1.0,
  "posterior_over_types": {type: probability, ...}
  (9 types, sum=1.0)}
```
- Belief only: same as probe but without "action" key

HISTORY FORMAT: t=round self_action opp_action
self_payoff opp_payoff

Listing 1: System prompt (Stage 1 & 2). <game_name>, <description>, <payoff_matrix> are filled at runtime per game.

A.2 Stage 1 — Action-Only User Prompt (Non-Probe Rounds)

Sent at every round $t \notin \{3, 6, 10\}$. History grows by ≈ 14 tokens per round.

```
Round <t>/<T>
t=01 C C 3.0 3.0
t=02 C D 0.0 5.0
t=03 D D 1.0 1.0
{"action": "C"} or {"action": "D"}
```

Listing 2: Action-only user prompt. History block is absent at $t = 1$.

A.3 Stage 1 — Belief Probe User Prompt (Probe Rounds)

A *separate* API call issued at $t \in \{3, 6, 10\}$ *after* the action call for that round. Does not affect the game state.

```
Round <t>/<T>
t=01 C C 3.0 3.0
t=02 C D 0.0 5.0
t=03 D D 1.0 1.0
[BELIEF PROBE] Reply with: "action",
"predicted_next_action", "confidence",
"posterior_over_types" (9 types, sum=1.0)
```

Listing 3: Belief probe user prompt (Stage 1, probe rounds $t \in \{3, 6, 10\}$).

Expected response (PROBE_SCHEMA):

```
{
  "action": "C",
  "predicted_next_action": "C",
  "confidence": 0.82,
  "posterior_over_types": {
    "always_cooperate": 0.35,
    "always_defect": 0.02,
    "tit_for_tat": 0.28,
```

```

    "grim_trigger":      0.15,
    "win_stay_lose_shift": 0.08,
    "noisy_tit_for_tat": 0.05,
    "deceptive_opportunist": 0.04,
    "gradual_defector":  0.02,
    "adaptive_mirror":   0.01
  }
}

```

Listing 4: Probe response schema. All nine opponent type keys are required; values must sum to 1.0.

A.4 Stage 2 — B2 Two-Stage Prompts

Call 1 (Belief-only):

```

Round <t>/<T>
t=01 C C 3.0 3.0
t=02 C D 0.0 5.0
[BELIEF ONLY] Reply with: "predicted_next_action",
"confidence", "posterior_over_types"
(9 types, sum=1.0). No action key.

```

Listing 5: B2 Call 1: belief elicitation without action. Used at probe rounds in Stage 2.

Call 2 (Action-from-posterior): The top-5 types from Call 1’s posterior are injected.

```

Round <t>/<T>
t=01 C C 3.0 3.0
t=02 C D 0.0 5.0
Your beliefs: {"tit_for_tat":0.412, "always_cooperate":0.281,
"grim_trigger":0.143, "adaptive_mirror":0.089,
"gradual_defector":0.075}
Choose action maximizing expected payoff.
{"action":"C"} or {"action":"D"}

```

Listing 6: B2 Call 2: action selection conditioned on the model’s own elicited posterior (top-5 shown).

A.5 Stage 3 — Oracle-Injected System and User Prompts

System prompt (Stage 3):

```

You are a strategic JSON-only agent playing a repeated
2x2 game.
Your entire response must be a single JSON object:
{"action": "C"} or {"action": "D"}

GAME: <game_name>
<description>

PAYOFF MATRIX:
You=C, Opponent=C -> you earn <cc0>, opponent earns <cc1>
You=C, Opponent=D -> you earn <cd0>, opponent earns <cd1>
You=D, Opponent=C -> you earn <dc0>, opponent earns <dc1>
You=D, Opponent=D -> you earn <dd0>, opponent earns <dd1>

You will be given a probability distribution over opponent
types. Each opponent type deterministically selects its next
action given the history. Choose the action that maximizes
expected immediate payoff under the provided posterior.

```

```
No explanation text.
```

Listing 7: Oracle-injected system prompt (Stage 3 / Experiment 3-C). Distinct from Stage 1 system prompt.

User prompt (Stage 3): The oracle posterior $P^*(\theta \mid h_t)$ is rendered as a top-5 probability table (4 decimal places).

```
Round <t>/<T>
t=01 C C 3.0 3.0
t=02 C D 0.0 5.0
Posterior over opponent types:
{"tit_for_tat":0.4821, "always_cooperate":0.2314,
"grim_trigger":0.1603, "adaptive_mirror":0.0712,
"gradual_defector":0.0550}
{"action":"C"} or {"action":"D"}
```

Listing 8: Oracle-injected user prompt. Oracle posterior is recomputed at every round from the true type.

A.6 Stage 4 — Posterior Intervention Prompts

System prompt (Stage 4):

```
You are a strategic JSON-only agent.
Your entire response must be:
{"action": "C"} or {"action": "D"}

GAME: <game_name>
<description>

PAYOFF MATRIX:
You=C, Opponent=C -> you earn <cc0>, opponent earns <cc1>
You=C, Opponent=D -> you earn <cd0>, opponent earns <cd1>
You=D, Opponent=C -> you earn <dc0>, opponent earns <dc1>
You=D, Opponent=D -> you earn <dd0>, opponent earns <dd1>

You will receive a probability distribution over opponent
types. Each type deterministically selects its next action.
Choose the action that maximizes expected immediate payoff
under the provided posterior.
No explanation text.
```

Listing 9: Intervention system prompt (Stage 4 / Experiment 4). No history in user prompt.

User prompt (Stage 4): No history is provided. The full 9-type posterior is serialised to 6 decimal places.

```
Round <t>/<T>
Posterior: {"always_cooperate": 0.010000,
"always_defect": 0.976250, "tit_for_tat": 0.001250,
"grim_trigger": 0.001250, "win_stay_lose_shift": 0.001250,
"noisy_tit_for_tat": 0.001250,
"deceptive_opportunist": 0.001250,
"gradual_defector": 0.001250,
"adaptive_mirror": 0.001250}
{"action":"C"} or {"action":"D"}
```

Listing 10: Intervention user prompt. Posterior only — no interaction history is included. This isolates the causal effect of the belief distribution on action selection.

A.7 JSON Schemas

Three schemas are used, enforced via structured output mode.

```
{
  "type": "object",
  "properties": {
    "action": {"type": "string", "enum": ["C", "D"]}
  },
  "required": ["action"],
  "additionalProperties": false
}
```

Listing 11: ACTION_SCHEMA: used for all non-probe action calls (Stages 1, 3, 4).

```
{
  "type": "object",
  "properties": {
    "action": {"type": "string", "enum": ["C", "D"]},
    "predicted_next_action": {"type": "string", "enum": ["C", "D"]},
    "confidence": {"type": "number"},
    "posterior_over_types": {
      "type": "object",
      "properties": {
        "always_cooperate": {"type": "number"},
        "always_defect": {"type": "number"},
        "tit_for_tat": {"type": "number"},
        "grim_trigger": {"type": "number"},
        "win_stay_lose_shift": {"type": "number"},
        "noisy_tit_for_tat": {"type": "number"},
        "deceptive_opportunist": {"type": "number"},
        "gradual_defector": {"type": "number"},
        "adaptive_mirror": {"type": "number"}
      },
      "required": ["always_cooperate", "always_defect",
        "tit_for_tat", "grim_trigger", "win_stay_lose_shift",
        "noisy_tit_for_tat", "deceptive_opportunist",
        "gradual_defector", "adaptive_mirror"],
      "additionalProperties": false
    },
  },
  "required": ["action", "predicted_next_action",
    "confidence", "posterior_over_types"],
  "additionalProperties": false
}
```

Listing 12: PROBE_SCHEMA: used for belief probe calls (Stage 1 probe rounds). All 9 type keys required.

```
{
  "type": "object",
  "properties": {
```



```

    "predicted_next_action": {"type": "string", "enum": ["C", "D"]},
    "confidence":             {"type": "number"},
    "posterior_over_types":   { ... }    // same as PROBE_SCHEMA
  },
  "required": ["predicted_next_action", "confidence",
               "posterior_over_types"],
  "additionalProperties": false
}

```

Listing 13: BELIEF_ONLY_SCHEMA: used for Stage 2 B2 Call 1. Identical to PROBE_SCHEMA minus the action field.

B Implementation Details

B.1 Model Configurations

Table 18 lists every provider key, exact model string, inference backend, and JSON enforcement mechanism used in this study. All evaluations use temperature = 0.0 and top- p = 1.0. Maximum token budgets are 512 for action calls and 1024 for probe calls.

Table 18: Full model configuration table. JSON Mode column indicates the mechanism used to enforce structured output.

Provider Key	Exact Model String	Backend	JSON Mode
openai	gpt-5.4-nano	OpenAI Chat Completions	json_schema (strict=true)
gemini	gemini-3.1-flash-lite-preview	Google GenAI SDK	response_schema + MIME
vertex_gemini	gemini-2.5-pro	Vertex AI (GCP)	response_schema + MIME
vertex_llama	meta/llama-4-maverick-17b-128e-instruct-maas	Vertex AI Model Garden	json_object
vertex_deepseek	deepseek-ai/deepseek-v3.2-maas	Vertex AI Model Garden	json_object
groq	llama-3.1-8b-instant	Groq (LL inference)	json_object

Provider-specific JSON enforcement. Three distinct mechanisms are used depending on provider capability:

- **OpenAI strict schema:** `response_format = {type: json_schema, json_schema: {strict: true, schema: ...}}`. Guarantees schema-valid output at the API level; eliminates parse failures for well-formed requests.
- **Anthropic prefill trick:** An assistant-role message pre-seeded with "{" is appended to the messages array, constraining the completion to begin inside a JSON object. The opening brace is prepended to the raw response before parsing.
- **Gemini response schema:** `response_mime_type = "application/json"` with `response_schema` set to the stripped schema. Fields `additionalProperties`, `$schema`, and `title` are removed as they are unsupported by the GenAI SDK. On schema field error, the request is retried without `response_schema` but with MIME-type enforcement retained.
- **json_object mode:** `response_format = {type: json_object}`. Enforces JSON syntax without field-level schema validation. Used for Groq and all Vertex Model Garden endpoints.

Gemini 2.5 Pro thinking mode. Gemini 2.5 Pro requires a minimum thinking budget. We set `thinking_budget=128` tokens and raise the maximum output budget to 4096 tokens for probe calls to prevent JSON truncation from competing with thinking tokens.

JSON parse fallback. All responses are first parsed with `json.loads()` on the full raw text. On `JSONDecodeError`, a fallback extracts the substring from the first `{` to the last `}` and re-attempts parsing. Failed parses return an empty dict `{}`, recorded as a parse failure. Schema failures (valid JSON but missing required fields) are tracked separately. Parse failures: 0 across all models and all experiments.

B.2 Rate-Limit and Retry Protocol

All API calls use exponential backoff with the schedule:

$$d_k = \min(1.0 \times 2^k, 30.0) \text{ seconds}, \quad k = 0, 1, \dots, 5$$

producing delays $\{1, 2, 4, 8, 16, 30\}$ seconds over at most $K = 6$ attempts. HTTP 429 and HTTP 5xx are retried; 4xx errors (excluding 429) raise immediately.

Table 19: Rate-limit defaults per provider. RPM = requests per minute; TPM = tokens per minute. Vertex AI providers use GCP project credit quotas with no hard RPM ceiling.

Provider	RPM	TPM	Base Delay (s)
openai	30	80,000	2.0
anthropic	30	80,000	2.0
groq	25	50,000	2.0
gemini	10	60,000	3.0
vertex_gemini	1,000	10,000,000	1.0
vertex_deepseek	1,000	10,000,000	1.0
vertex_llama	1,000	10,000,000	1.0

B.3 Checkpointing and Reproducibility

State is persisted after every completed condition via atomic write (`os.replace` on POSIX). Checkpoint files are named:

`ckpt_pipeline_{provider}_{model}_{tag}.json`

On restart, the pipeline loads `completed` and `pending` lists and resumes from the first pending entry. API log entries are stripped before serialization to bound file size; they are retained in memory during a run.

Episode seeds are assigned as:

$$s = s_0 + \text{hash}(\mathcal{G}, \theta, i) \bmod 2^{31}, \quad s_0 = 42$$

ensuring identical opponent sequences across all five stages within the same condition and full reproducibility across provider re-runs.

C Full Per-Game Per-Model Results

C.1 Experiment 1: Normalized Payoff and Cooperation Rate by Game

Table 20: Experiment 1: Normalized payoff (NP) and cooperation rate (CR) per model per game. 95% CI in brackets. $N = 90$ episodes per cell.

Model	Metric	PD	SH	Chicken	BoS
DeepSeek V3.2	NP	0.436 [0.389, 0.482]	0.868 [0.824, 0.911]	0.342 [0.282, 0.403]	0.889 [0.851, 0.927]
	CR	0.129 [0.077, 0.181]	0.797 [0.728, 0.866]	0.163 [0.108, 0.219]	0.832 [0.767, 0.897]
Gemini 2.5 Pro	NP	0.536 [0.503, 0.569]	0.871 [0.828, 0.914]	0.424 [0.373, 0.476]	0.895 [0.858, 0.931]
	CR	0.714 [0.657, 0.772]	0.810 [0.741, 0.879]	0.389 [0.351, 0.427]	0.837 [0.774, 0.900]
Gemini Flash Lite	NP	0.525 [0.496, 0.555]	0.881 [0.839, 0.923]	0.329 [0.293, 0.365]	0.899 [0.862, 0.936]
	CR	0.807 [0.738, 0.875]	0.809 [0.740, 0.878]	0.399 [0.352, 0.446]	0.811 [0.742, 0.880]
LLaMA 3.1 8B	NP	0.507 [0.476, 0.539]	0.886 [0.845, 0.926]	0.486 [0.444, 0.527]	0.759 [0.739, 0.779]
	CR	0.782 [0.710, 0.854]	0.817 [0.750, 0.883]	0.793 [0.723, 0.864]	0.598 [0.556, 0.640]
GPT-5.4 Nano	NP	0.520 [0.489, 0.550]	0.875 [0.833, 0.917]	0.523 [0.493, 0.553]	0.865 [0.818, 0.913]
	CR	0.848 [0.790, 0.906]	0.838 [0.775, 0.901]	0.932 [0.905, 0.960]	0.900 [0.859, 0.941]
LLaMA 4 Maverick	NP	0.516 [0.485, 0.546]	0.876 [0.833, 0.918]	0.444 [0.406, 0.482]	0.891 [0.854, 0.928]
	CR	0.801 [0.733, 0.869]	0.799 [0.730, 0.868]	0.492 [0.475, 0.510]	0.802 [0.734, 0.871]

C.2 Experiment 4: Directional Accuracy by Game

Table 21: Experiment 4: Action-flip-valid directional accuracy ($\text{DirAcc}_{\text{flip}}$) and decision sensitivity (Sens) by game. PD has no flip-valid probes (dominant strategy); margin-only DirAcc is reported instead. $N =$ valid probes per game.

Model	PD (margin only)		Stag Hunt		Chicken		BoS	
	DirAcc	Sens	DirAcc	Sens	DirAcc	Sens	DirAcc	Sens
DeepSeek V3.2	0.670	0.670	0.981	0.509	0.150	0.527	0.981	0.488
Gemini 2.5 Pro	0.758	0.758	0.978	0.472	0.859	0.508	0.996	0.490
Gemini Flash Lite	0.596	0.596	0.985	0.482	0.812	0.492	0.996	0.482
LLaMA 3.1 8B	0.151	0.151	1.000	0.315	0.067	0.082	0.913	0.495
GPT-5.4 Nano	0.812	0.812	0.996	0.484	0.129	0.347	0.964	0.529
LLaMA 4 Maverick	0.363	0.363	1.000	0.490	0.253	0.076	1.000	0.482

C.3 Experiment 5: Cognitive Theory Metrics by Game

Table 22: Experiment 5: JS divergence (H1), EU-coherence (H2), ECE (H4), and deceptive-type gap (H6) per model per game. RI (H5) is a single point estimate across 4 games (shown in master table). H7 = 0.000 for all models and games (omitted).

Model	Metric	PD	SH	Chicken	BoS
DeepSeek V3.2	JS	0.381	0.217	0.371	0.290
	Coh	0.889	0.948	0.844	0.967
	ECE	0.217	0.187	0.264	0.174
	H6 Gap	0.345	0.126	0.362	0.145
Gemini 2.5 Pro	JS	0.294	0.228	0.432	0.241
	Coh	0.467	0.930	0.822	0.967
	ECE	0.079	0.062	0.295	0.103
	H6 Gap	0.224	0.136	0.400	0.286
Gemini Flash Lite	JS	0.247	0.233	0.399	0.228
	Coh	0.215	0.982	0.452	0.970
	ECE	0.149	0.124	0.298	0.157
	H6 Gap	0.102	0.119	0.138	0.086
LLaMA 3.1 8B	JS	0.513	0.521	0.525	0.547
	Coh	0.067	0.833	0.182	0.589
	ECE	0.172	0.119	0.119	0.158
	H6 Gap	0.167	0.143	0.143	0.143
GPT-5.4 Nano	JS	0.166	0.157	0.162	0.156
	Coh	0.204	0.993	0.089	0.985
	ECE	0.152	0.138	0.188	0.143
	H6 Gap	0.291	0.295	0.291	0.291
LLaMA 4 Maverick	JS	0.252	0.277	0.275	0.300
	Coh	0.226	0.974	0.589	0.978
	ECE	0.184	0.145	0.318	0.173
	H6 Gap	0.286	0.286	0.343	0.291

Table 17: Experiment 5: Per-game cognitive breakdown. JS (H1), Coh (H2), ECE (H4), and H6 deceptive-type gap.

Model	Metric	PD	SH	Chicken	BoS
DeepSeek V3.2	JS	0.381	0.217	0.371	0.290
	Coh	0.889	0.948	0.844	0.967
	ECE	0.217	0.187	0.264	0.174
	H6 Gap	0.345	0.126	0.362	0.145
Gemini 2.5 Pro	JS	0.294	0.228	0.432	0.241
	Coh	0.467	0.930	0.822	0.967
	ECE	0.079	0.062	0.295	0.103
	H6 Gap	0.224	0.136	0.400	0.286
Gemini Flash Lite	JS	0.247	0.233	0.399	0.228
	Coh	0.215	0.982	0.452	0.970
	ECE	0.149	0.124	0.298	0.157
	H6 Gap	0.102	0.119	0.138	0.086
LLaMA 3.1 8B	JS	0.513	0.521	0.525	0.547
	Coh	0.067	0.833	0.182	0.589
	ECE	0.172	0.119	0.119	0.158
	H6 Gap	0.167	0.143	0.143	0.143
GPT-5.4 Nano	JS	0.166	0.157	0.162	0.156
	Coh	0.204	0.993	0.089	0.985
	ECE	0.152	0.138	0.188	0.143
	H6 Gap	0.291	0.295	0.291	0.291
LLaMA 4 Maverick	JS	0.252	0.277	0.275	0.300
	Coh	0.226	0.974	0.589	0.978
	ECE	0.184	0.145	0.318	0.173
	H6 Gap	0.286	0.286	0.343	0.291